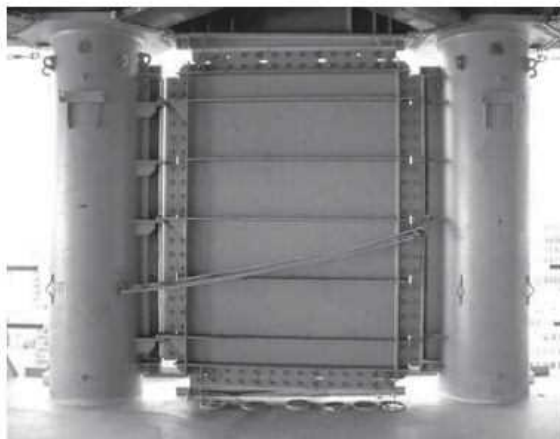


طراحی و آنالیز دیوار برشی فولادی

ایمان الیاسیان، کارشناس ارشد سازه i.elyasian@gmail.com

- در طراحی دیوار برشی فولادی شبیه به تیر ورق تئوریهای مختلف توسط محققان ارائه شده است که در زیر به برخی از آنها اشاره می گردد ۱- تئوری باسلر Basler در ۱۹۶۱ (جان ضخیم و بالهای تیر انعطاف پذیر) بنابراین نیروهای ناشی از کششی قطری پس از کمانش برش جان توسط بالها به دلیل انعطاف پذیر بودن تحمل می شود ۲- Takeucgi-1964 (ناحیه تسلیم)
- ۳- Fuji 1968-1971 (ناحیه تسلیم) ۴- Kumatso 1971 (ناحیه تسلیم) بایستی توجه داشت ناحیه تسلیم تئوری Chen and Ostapenko 1969 و Fuji , Kumatso باهم متفاوت هستند ۵- Porter (تشکیل مفصل پلاستیک) Hogland 1971-۷ برای اصل استوار است که تنشهای غشایی در جهت عمود بر پانل جان وجود ندارد و برای تیورقهای طویل بدون سخت کننده های میانی صحیح می باشد تنشهای فشاری غشایی پس از کمانش افزایش نمی یابد ولی تنشهای غشایی کششی تا زمان مقاومت نهایی افزایش می یابد ۸- تئوری تسلیم Herzog 1974
- ۹- تیوری تسلیم اشنایدر و شرودر Steinhardt-Schroder 1971-۱۰ تیوری کلارک و شارپ برای تیورقهای آلومینیومی با جان نازک ارائه شده است و شامل میدان کششی تئوری باسلر و میدان کششی با زاویه ۴۵ درجه در کل جان گسترده شده ۱- شکست خمشی در پانلهای میانی و شکست برش در پانلهای کناری اتفاق می افتد.
- ۲- انواع از نظر باری
- ۳- دیوار تحت بار جانبی در جهت ضعیف
- ۴- دیوار تحت بار قائم ۳- دیوار تحت بار جانبی جهت قوی
- ۵- دیوار تحت بار جانبی در جهت قوی و بار قائم
- ۶- انواع دیوار بتن آرمه از نظر رفتاری
- ۷- ۱- Panel wall (دیوار محیطی متصل به قاب) ۲- Curtain wall (دیوار محیطی مهارشده به قاب)
- ۸- ۳- Partition wall ۴- دیوار حائل Retaining wall ۵- دیوار زیر زمینی basement wall
- ۹- ۶- دیوار برشی shear wall

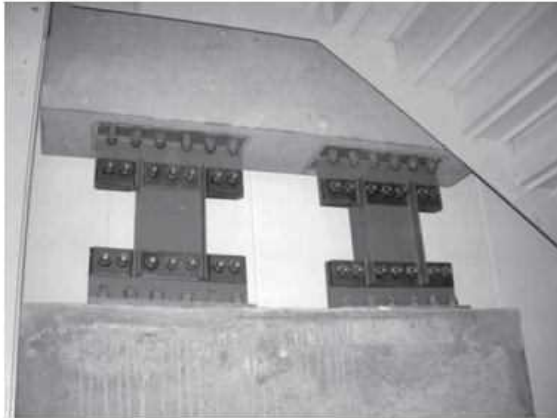


(a)

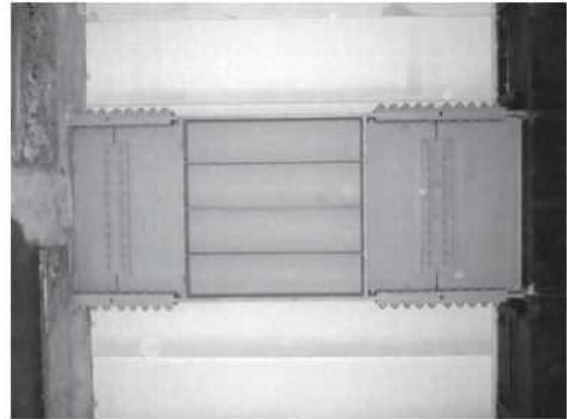


(b)

SPW panel in Japan: (a) wall with horizontal panel stiffeners (courtesy of Takanaka);
(b) wall with horizontal and vertical stiffeners (courtesy of Nippon Steel).



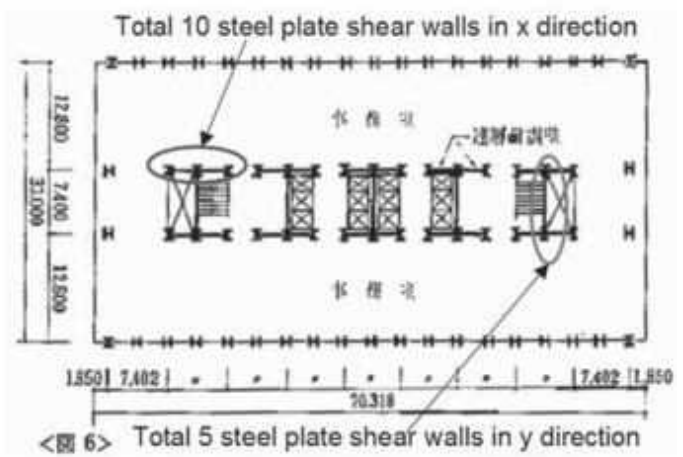
*Small shear yielding elements in Japan
(courtesy of Shimizu).*



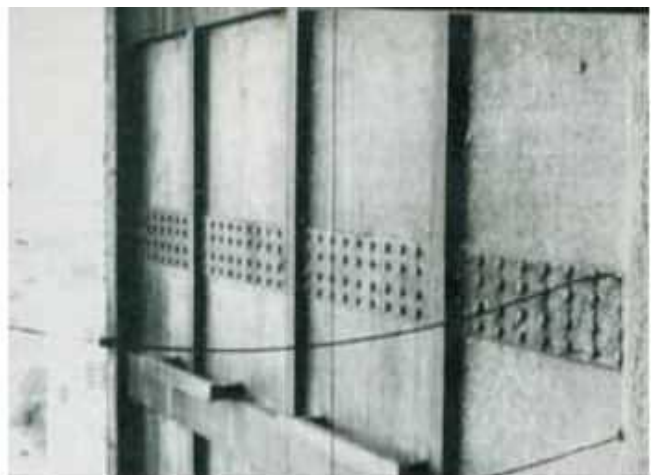
*Shear link connected between closely spaced columns
(courtesy of Nippon Steel).*

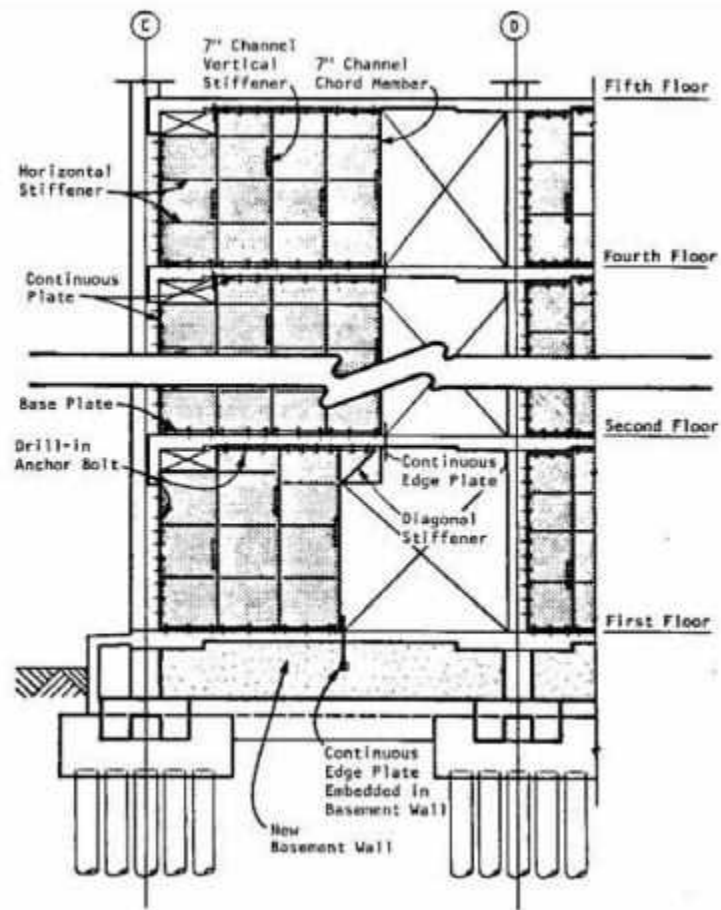


*Intermediate column approach used in Japan
(courtesy of Takanaka).*



*Plan layout of SPW for Nippon Steel Building
(Yokoyama et al., 1978).*



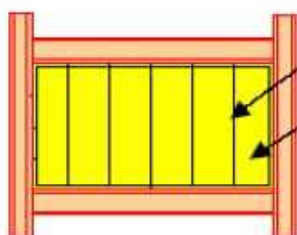




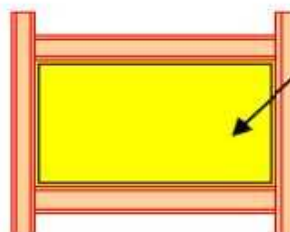
SPW in Canam Manac Group headquarters expansion
(courtesy of Richard Vincent, Canam Manac Group,
St. George, Quebec, Canada).



Close-up view of wall at mid-span splice location, ING
building (courtesy of Louis Crepeau and Jean-Benoit Ducharme,
Groupe Teknika, Montreal, Canada).

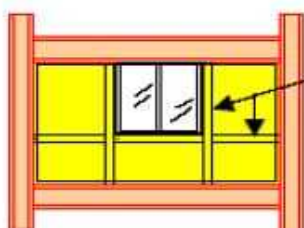


Stiffeners on the Front Face
of the Wall
Stiffeners on the Back
Side of the Wall

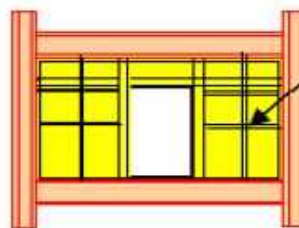


Steel Plate
Shear Wall

دیوار برشی فولادی با سخت کننده و بدون آن



Boundary
Members



Stiffeners

دیوار با سخت کننده و بازشو

دیوار با سخت کننده های دوطرفه

دیوار با سخت کننده و با بازشو

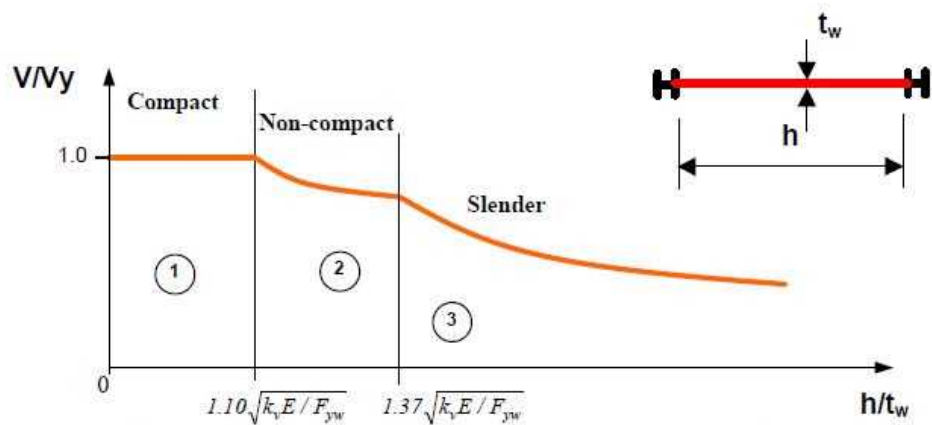
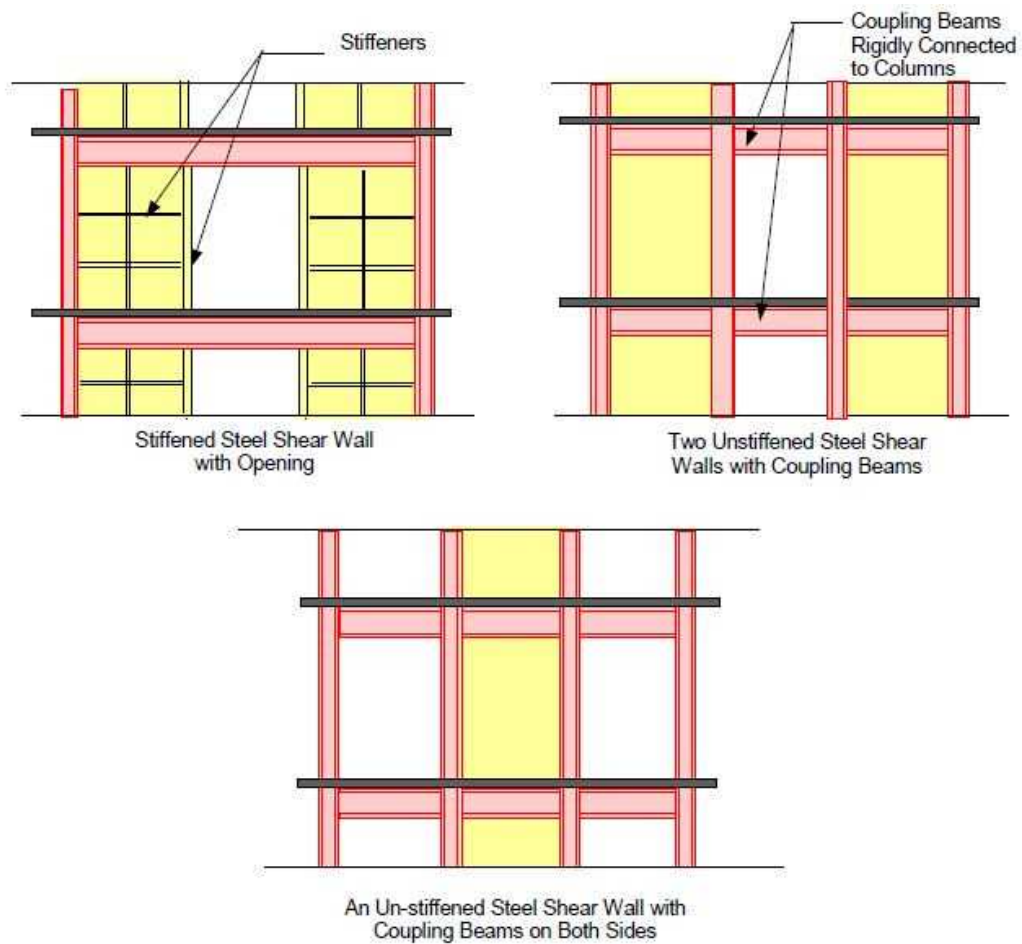
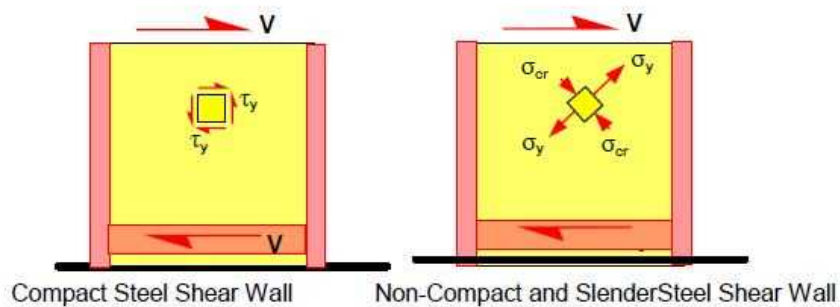
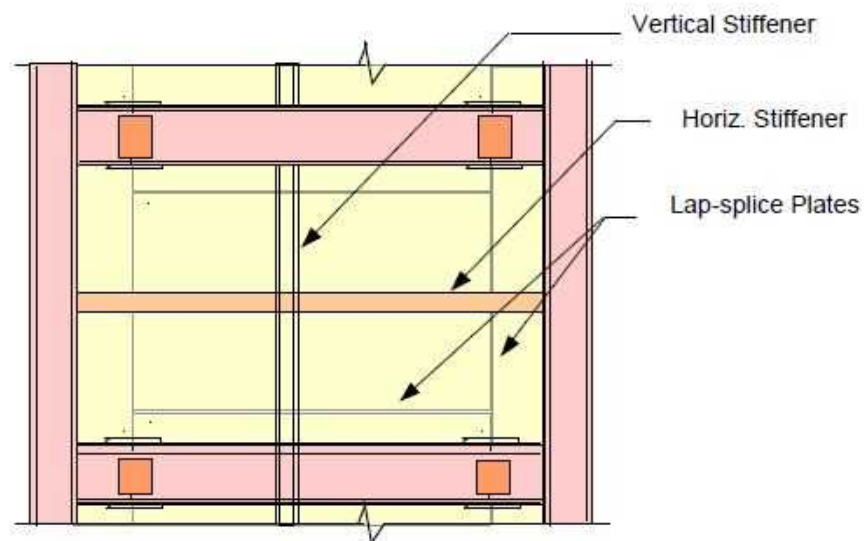


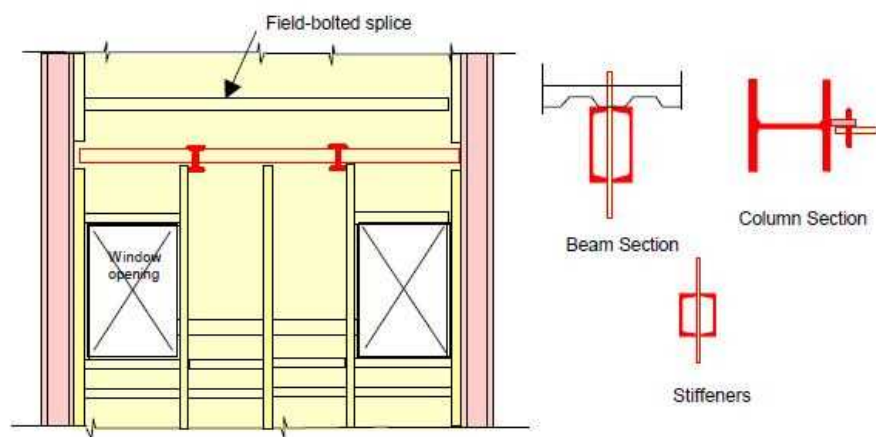
Figure 5.2. Three regions of behavior of steel shear walls



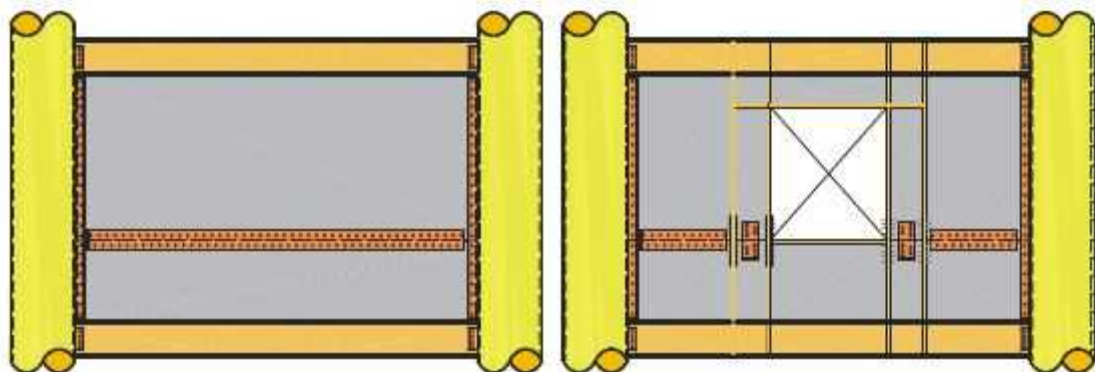
Steel shear walls resisting shear in “shear yielding” and tension field action



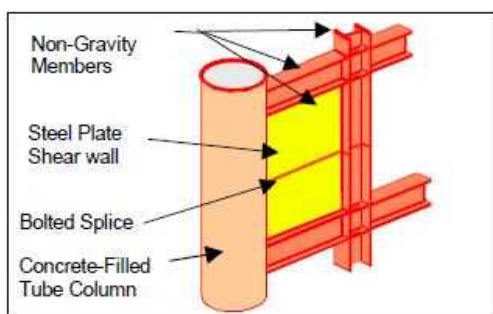
Details of steel shear walls used in Nippon Steel Building



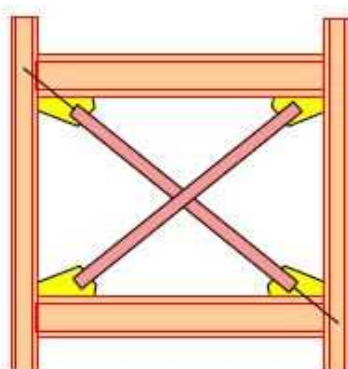
A view of stiffened steel plate shear walls of Sylmar hospital



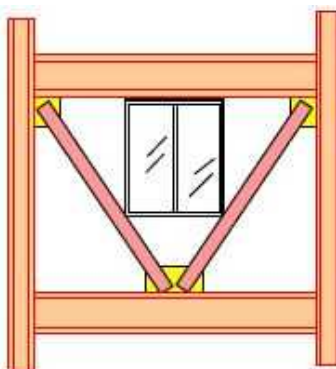
A typical SPSW bay with and without opening (after BPA Group)



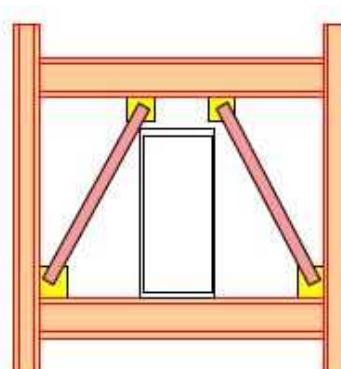
Components of the tested system and bolted splice



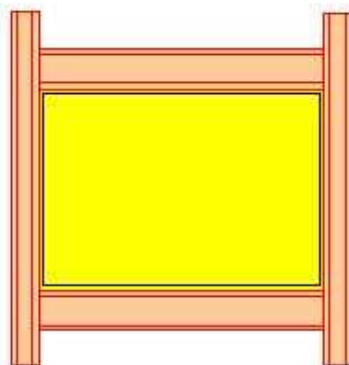
X-bracing



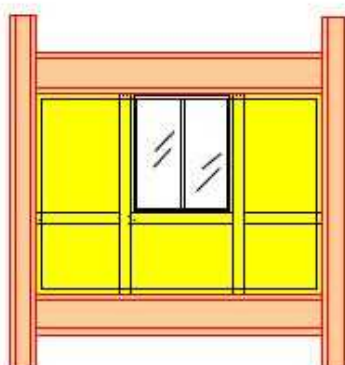
V-bracing



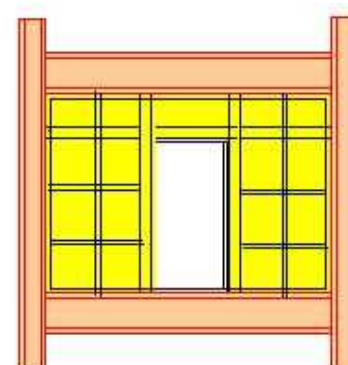
Eccentric Bracing



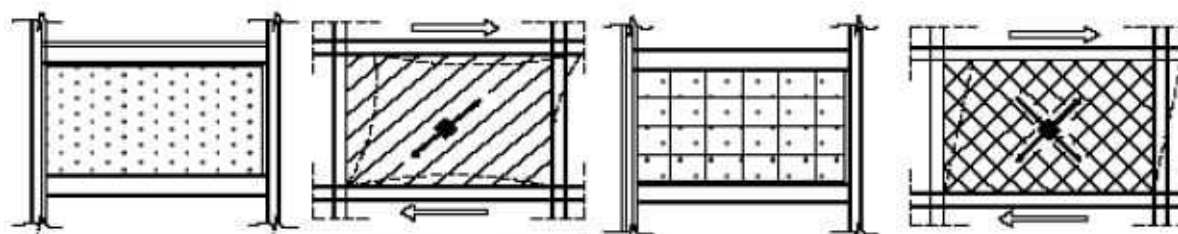
Steel Plate Shear Wall
(Unstiffened)



Stiffened Steel Shear
Wall With Opening



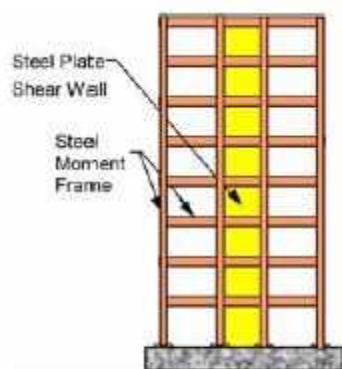
Stiffened Steel Shear
Wall with Opening



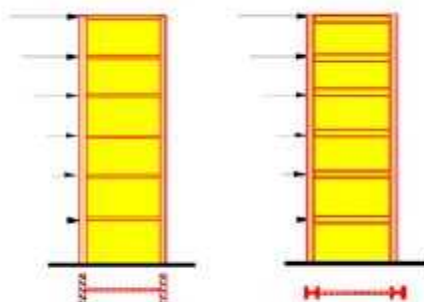
مکانیزم میدان کششی (پانل سخت نشده)

مکانیزم برشی خالص (پانل سخت شده)

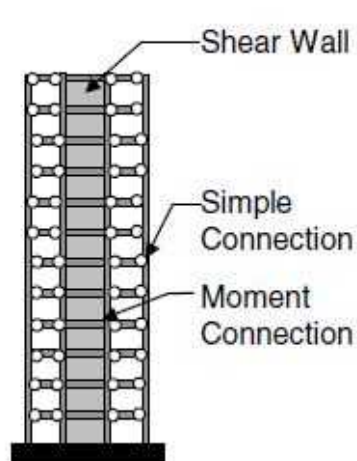
مکانیزمهای میدان کششی و برشی خالص



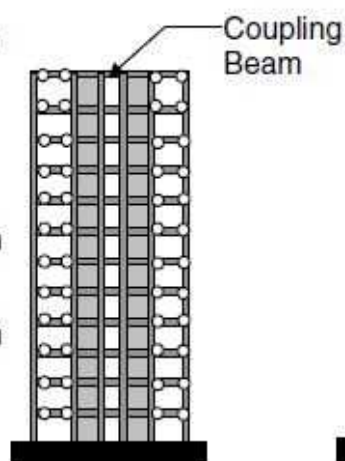
دیوار برشی فولادی و قاب خمشی محیطی



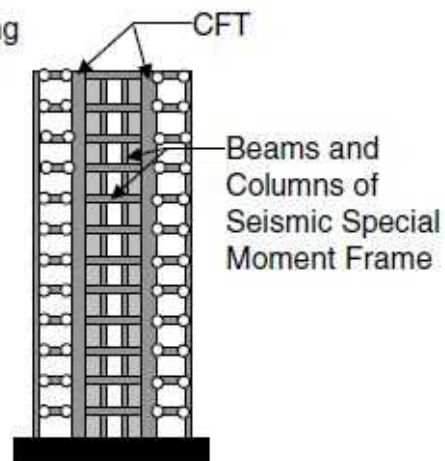
دیوار برشی فولادی سمت راست و تیر ورق سمت چپ



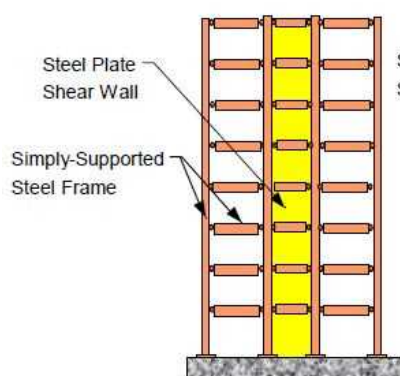
Type I



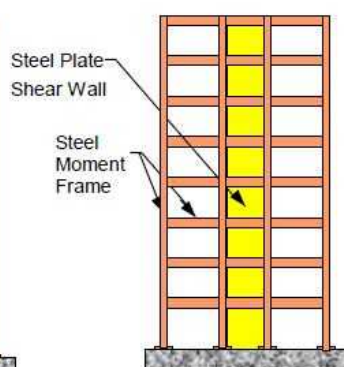
Type II



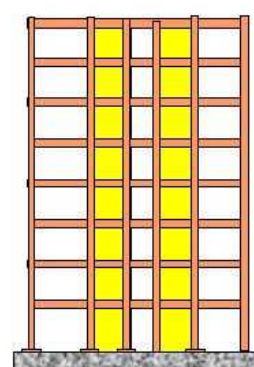
Type III



a. Singular Shear Wall Inside Gravity Frame

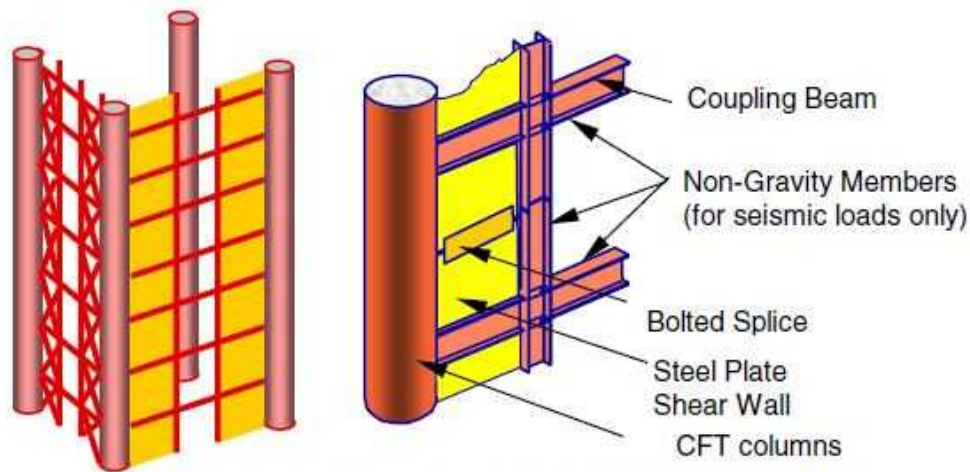


b. Dual System with Shear Wall Inside Moment Frame

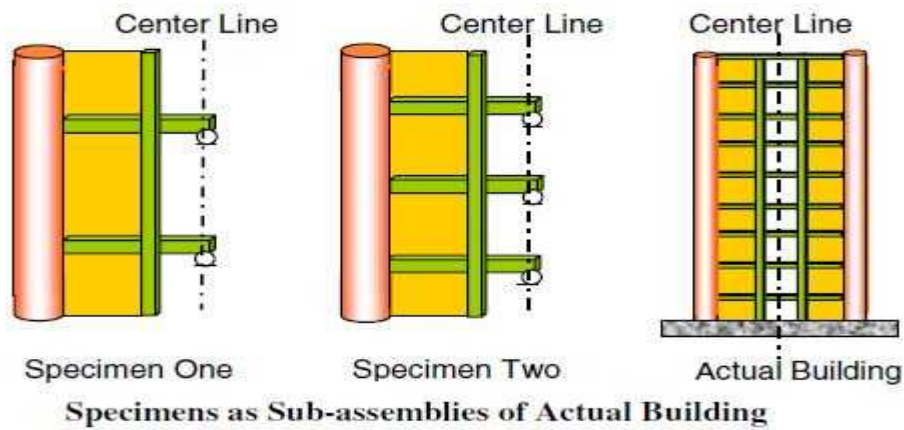


c. Dual System with Coupling Beams

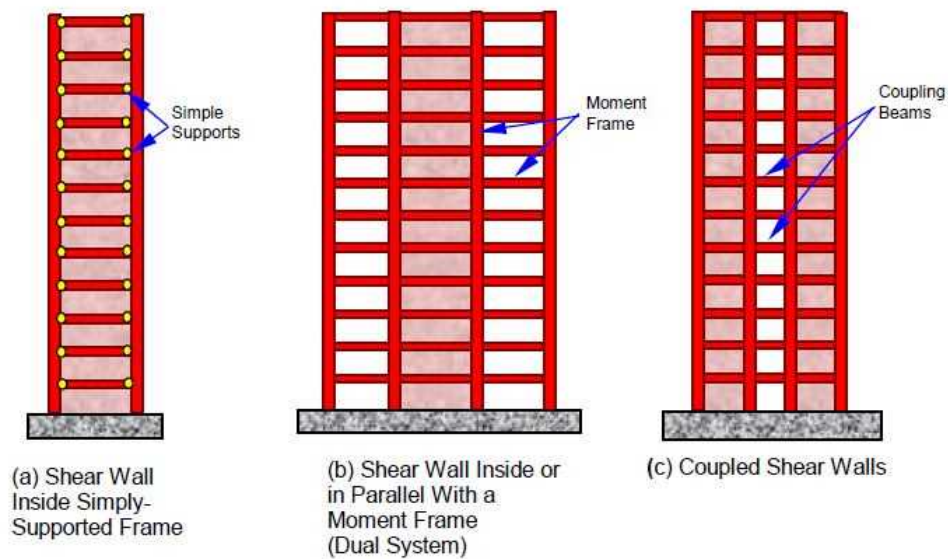
Typical Steel Shear Wall Systems



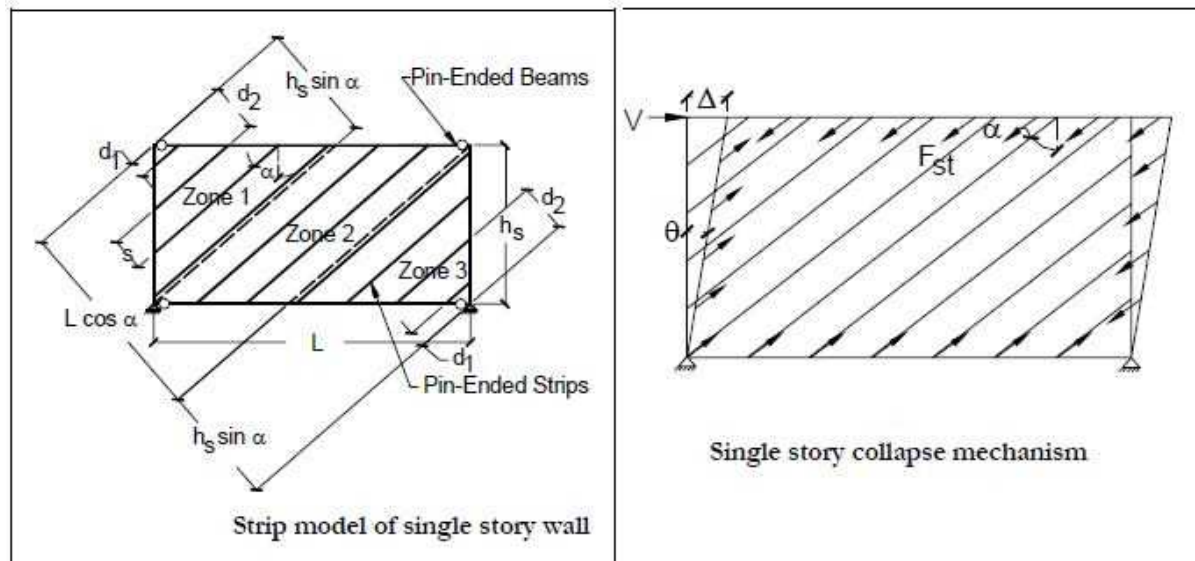
Lateral Load Resisting System of a Steel Building



Specimens as Sub-assemblies of Actual Building

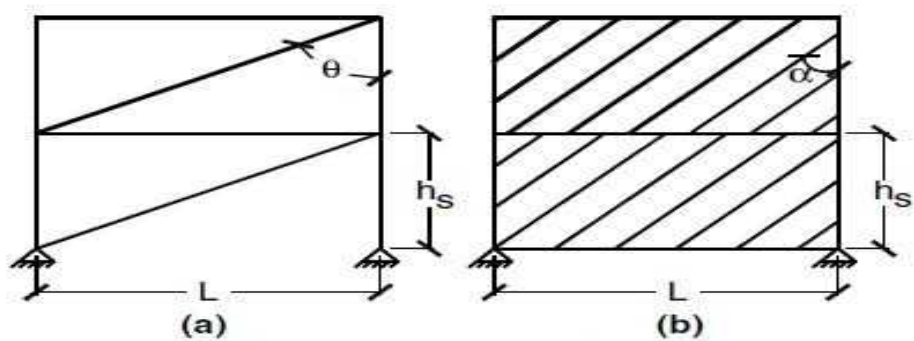


Typical Steel Structure with Composite Shear Walls

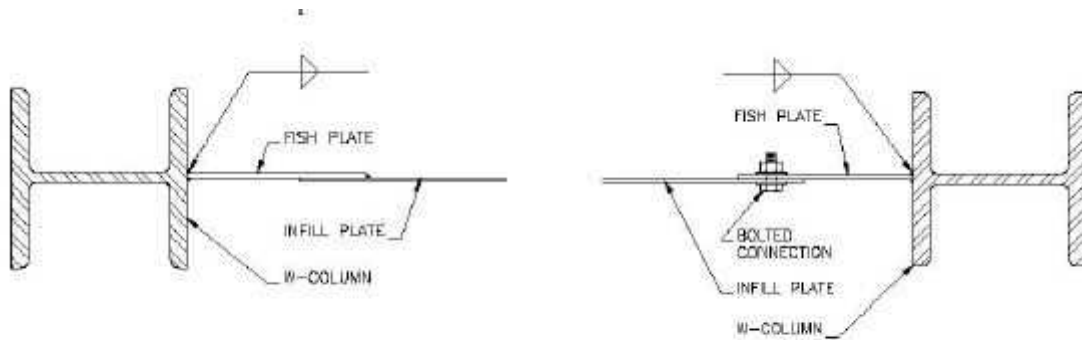


$$V = \frac{1}{2} F_y t L \sin(2\alpha) + \frac{4 M_{pc}}{h_s}$$

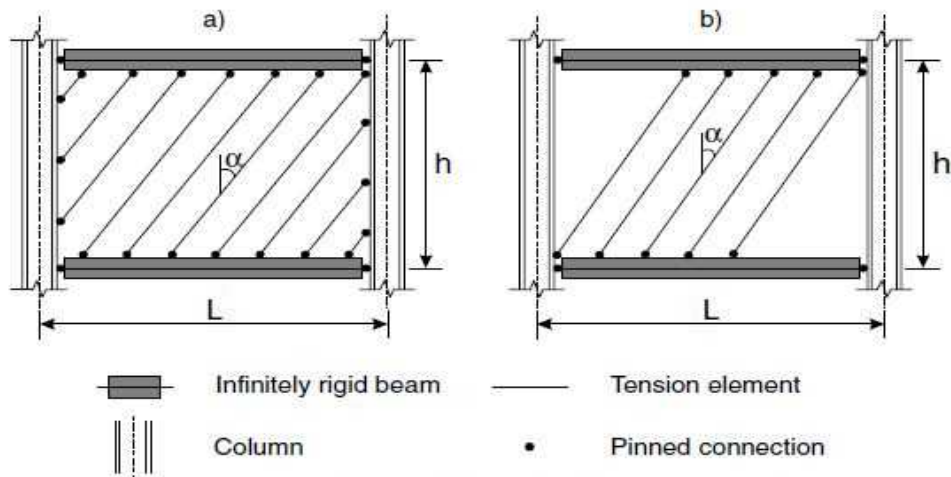
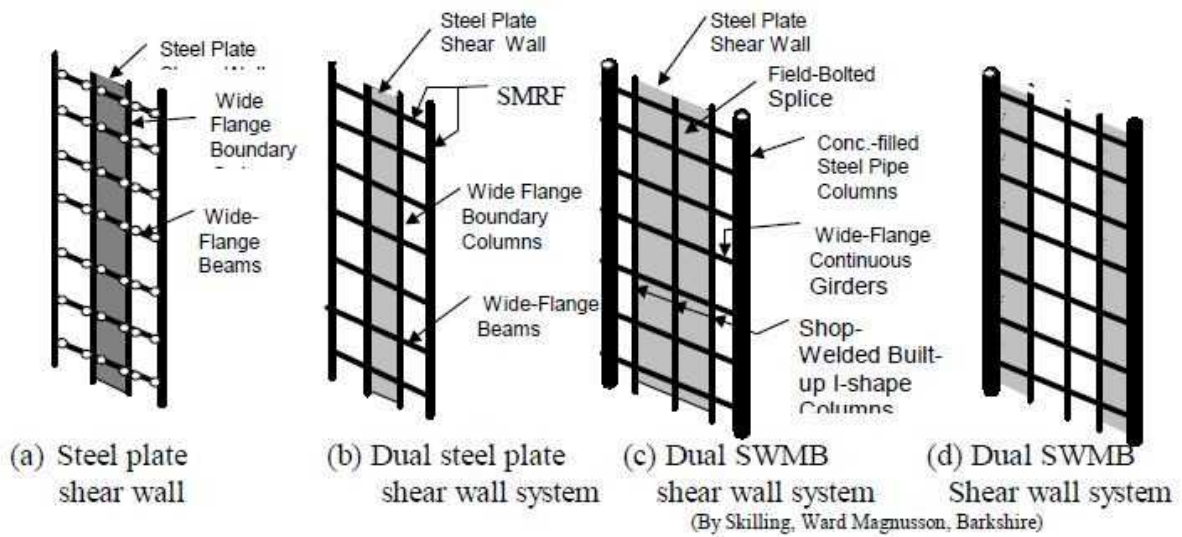
$$t = \frac{2A \sin \theta \sin 2\theta}{L \sin^2 2\alpha} \quad \tan^4 \alpha = \frac{1 + \frac{tL}{2A_c}}{1 + t h_s \left(\frac{1}{A_b} + \frac{h_s^3}{360 I_c L} \right)} \quad I_c \geq \frac{0.00307 t h_s^4}{L}$$



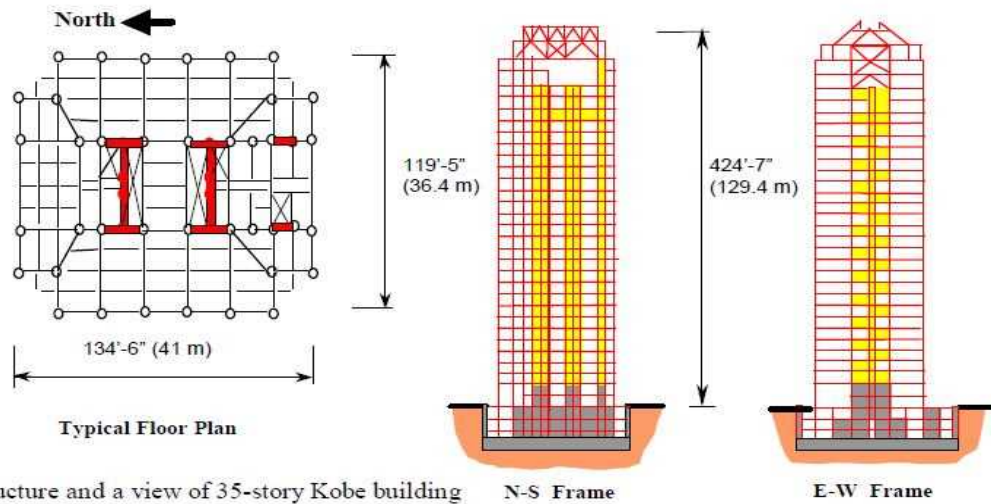
(a) Equivalent Story Brace Model
(b) The Strip Model



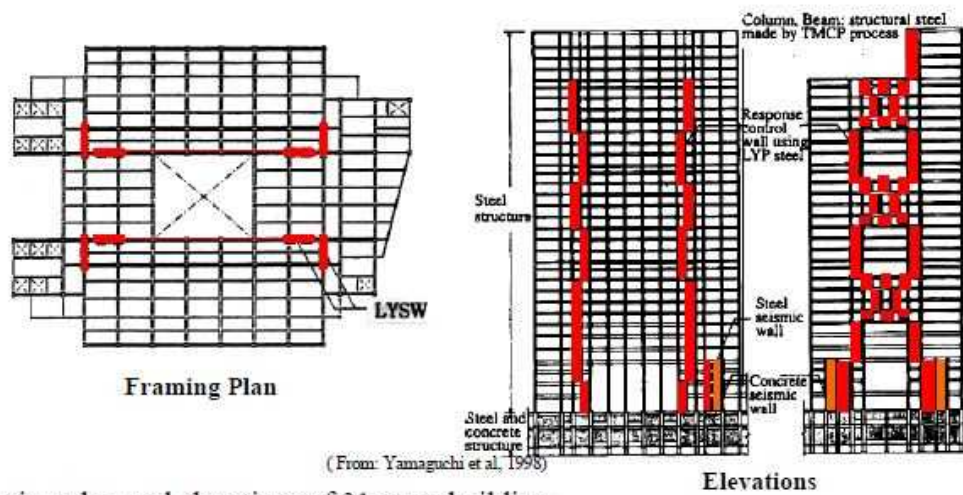
Typical "fish plate" to boundary frame and infill plate to "fish plate" connections



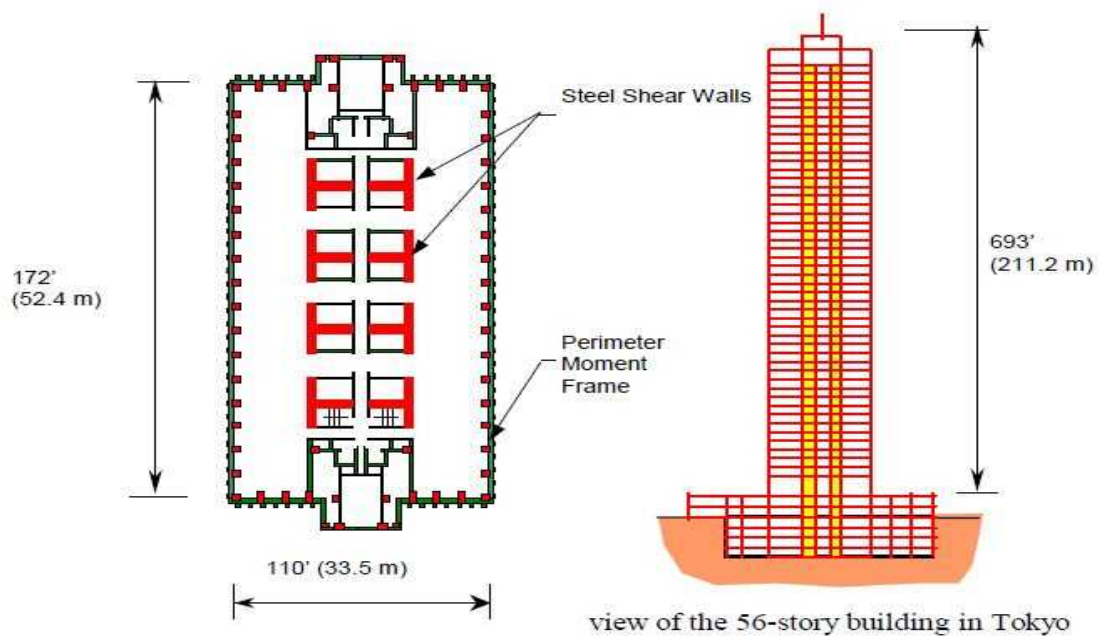
Strip model representation of a typical steel panel developed by Thorburn et al. (1983) a) complete tension field b) partial tension field

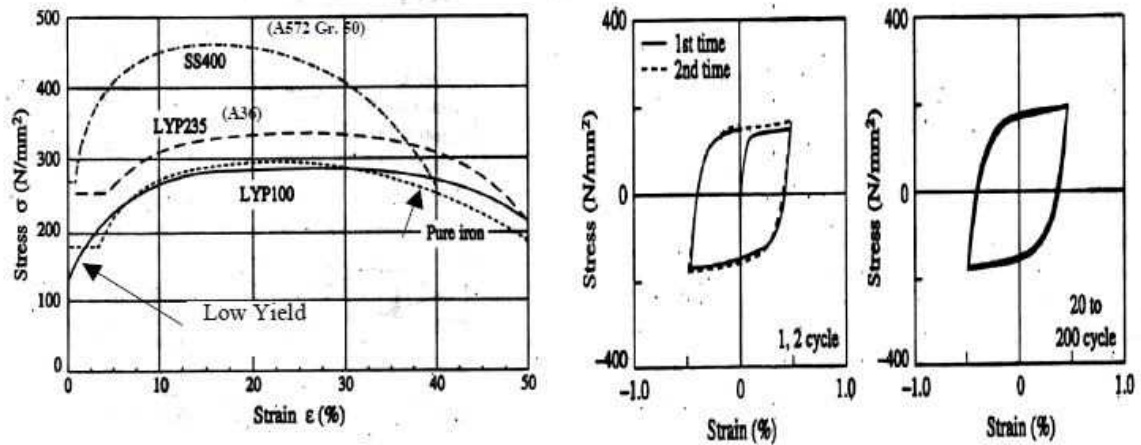


Structure and a view of 35-story Kobe building



Framing plan and elevations of 31-story building in Japan using low-yield steel shear walls

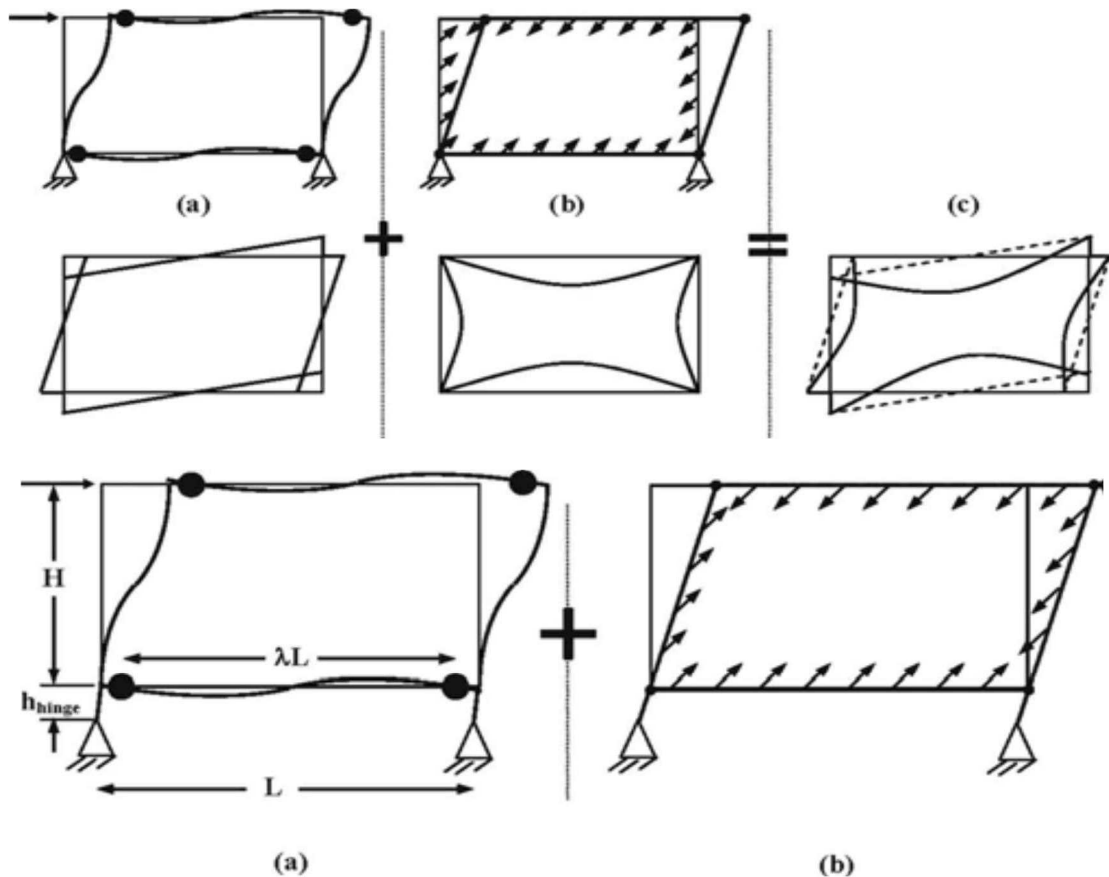


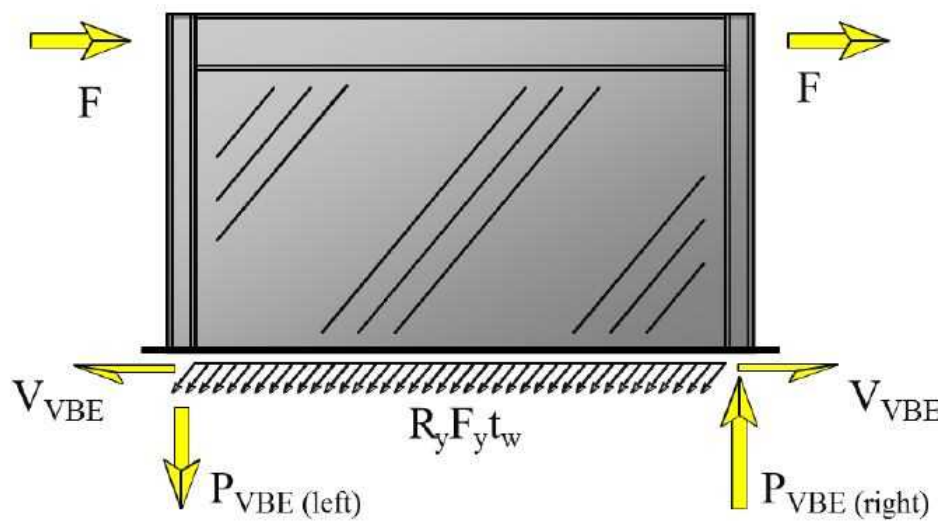
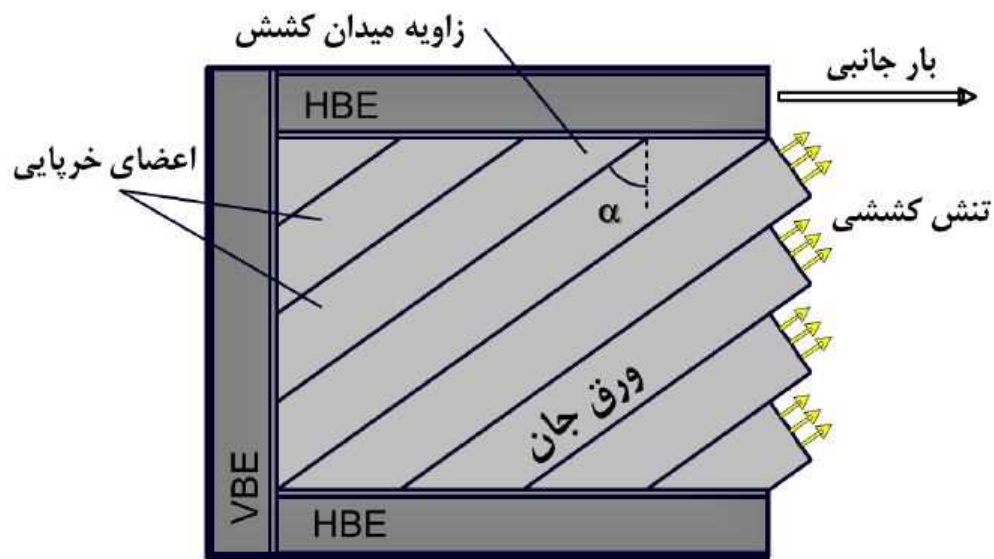


Stress-strain curves and hysteresis behavior of low-yield steel
(Nippon Steel, 1998)

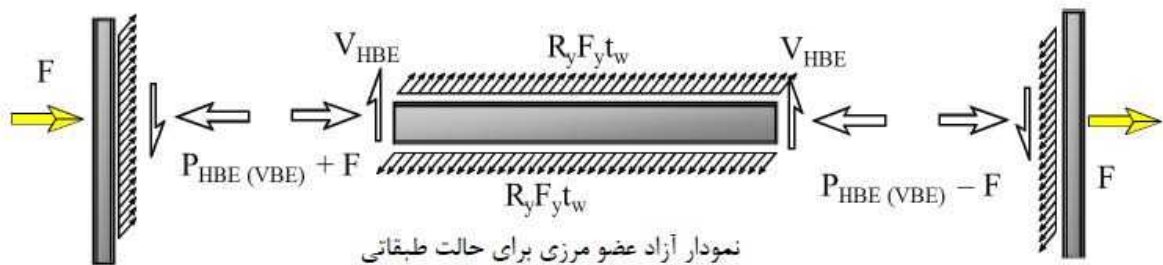
Steel Plate Shear Wall Systems in Canadian Code and Their R-Factors

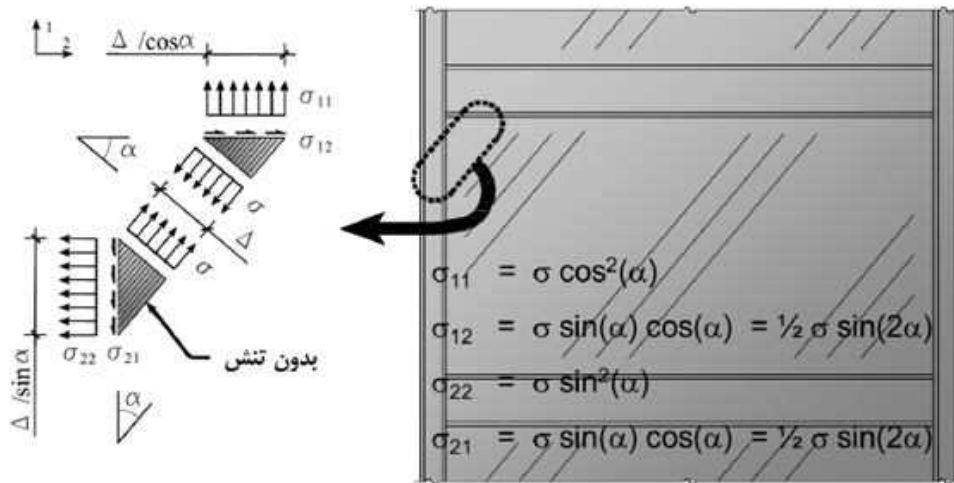
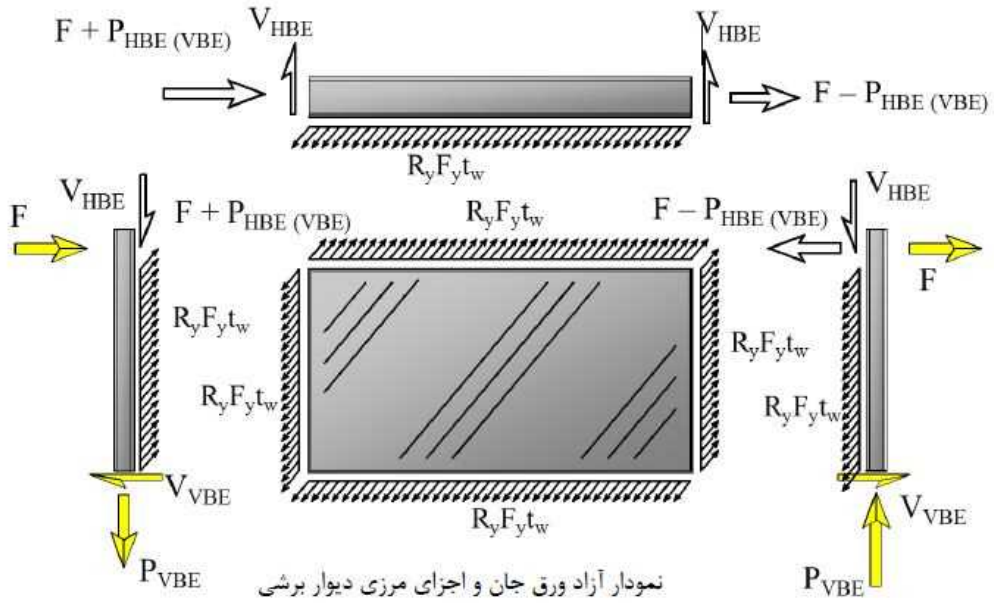
Type of Steel Plate Shear Wall	Requirement	R_{CAN} (Canadian)	R_{US} (US Equivalent)
Ductile	The frame containing the wall should be ductile moment frame	4.0	8.0
Nominally Ductile	The frame containing the wall should be nominally ductile moment frame (Intermediate in the U.S. definition).....	3.0	6.0
Ordinary	No specific requirement for frame. It can be a frame with pin connections.	2.0	4.0





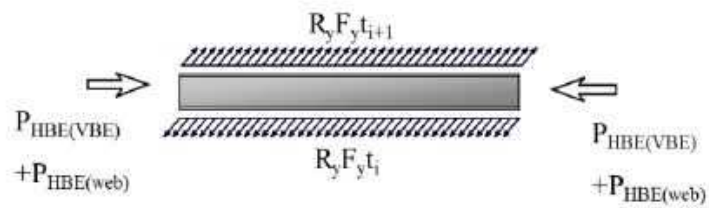
نیروها اعمالی به دیوار برشی فولادی همراه به عکس العمل‌های تکیه‌گاهی





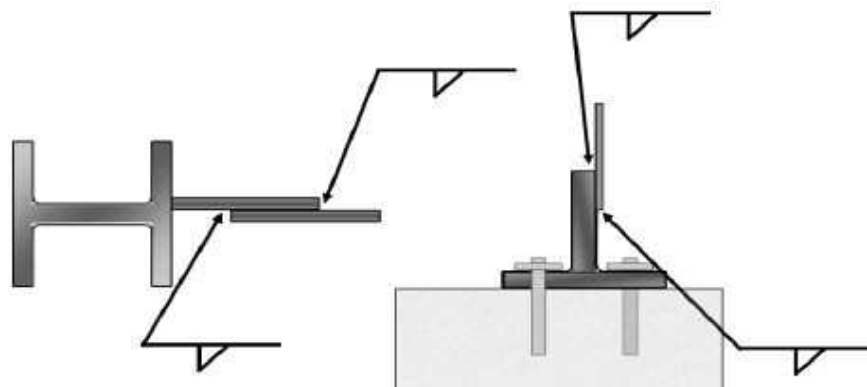
$$P_{HBE(VBE)} = \sum \frac{1}{2} R_y F_y t_w h_c \sin^2 \alpha$$

$$P_{HBE(web)} = \frac{1}{2} R_y F_y [t_i \sin(2\alpha_i) - t_{i+1} \sin(2\alpha_{i+1})] L_{cf}$$

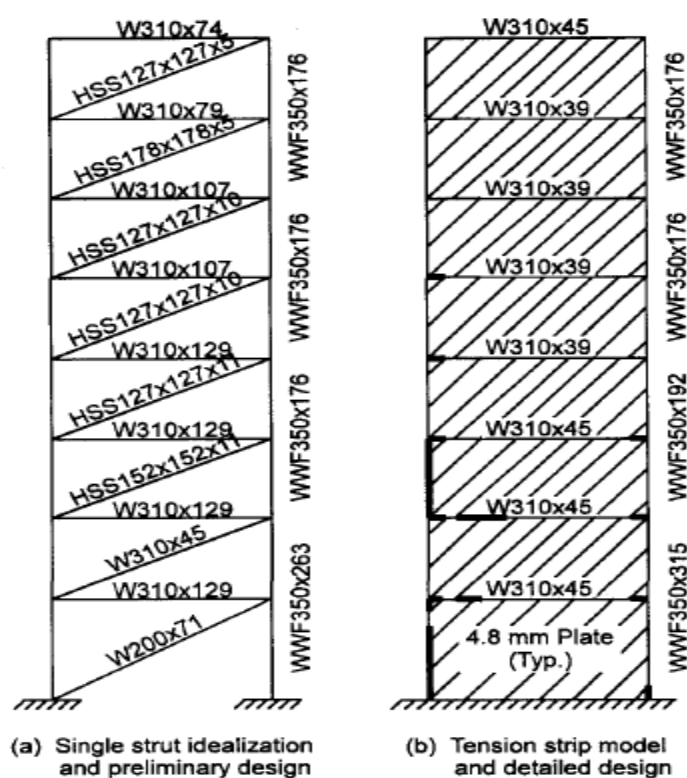


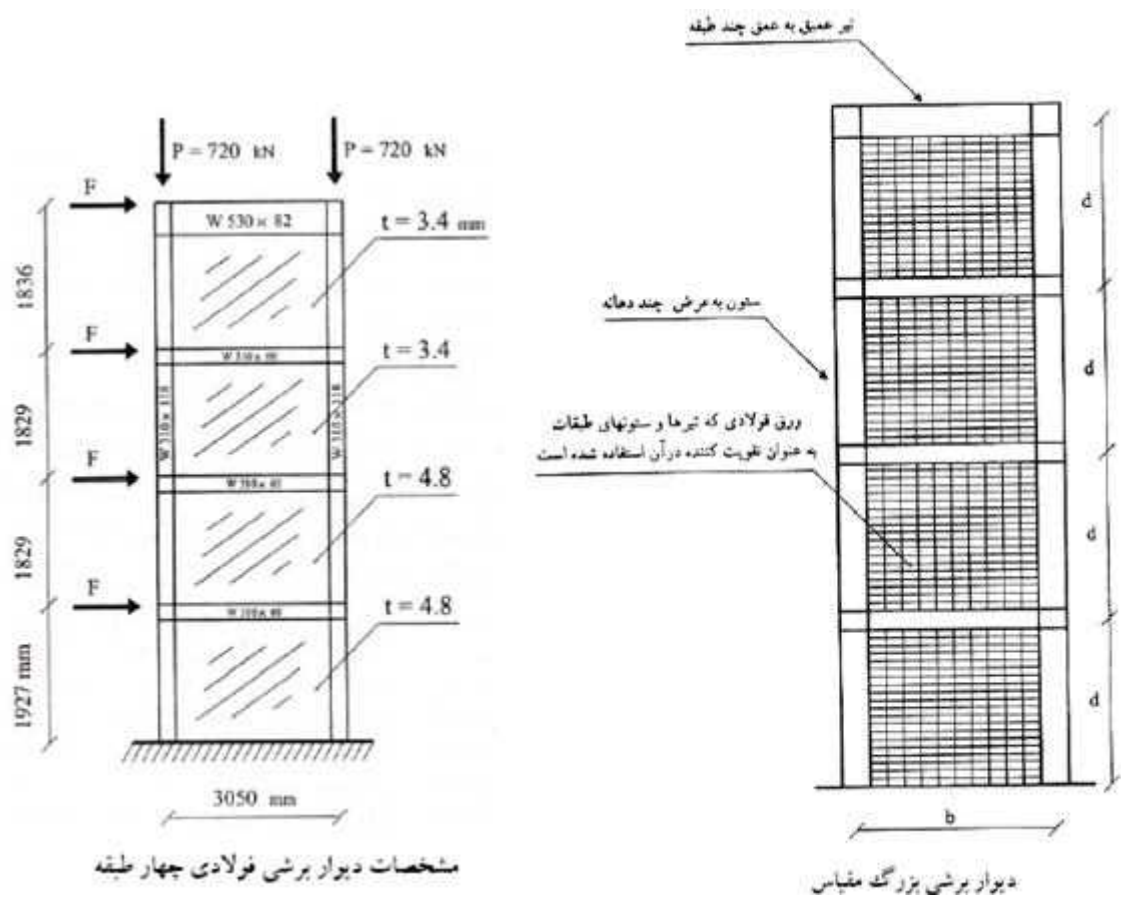
$$V_n = 0.42 F_y t_w L_{cf} \sin 2\alpha$$

$$I_c \geq 0.00307 \frac{t_w h^4}{L}$$

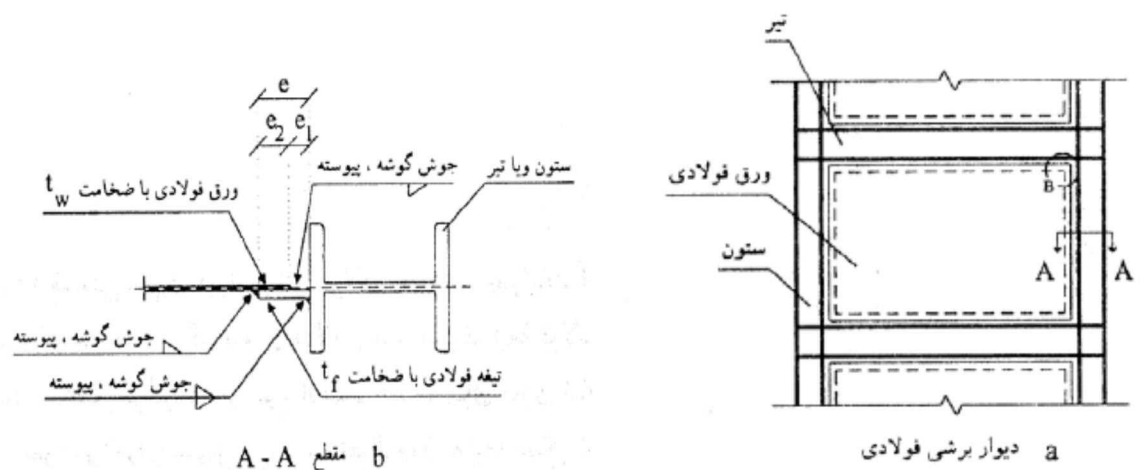


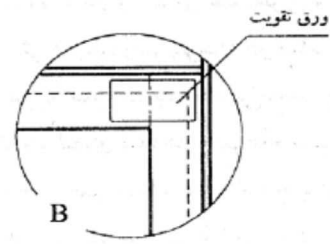
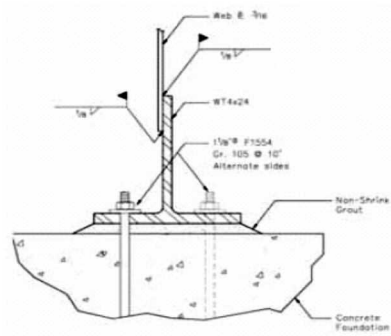
جزئیات اتصال ورق جان به اجزای مرزی



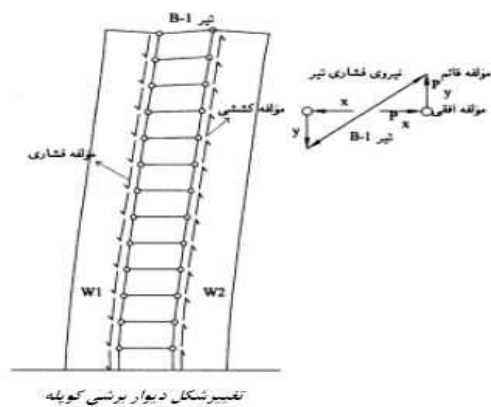
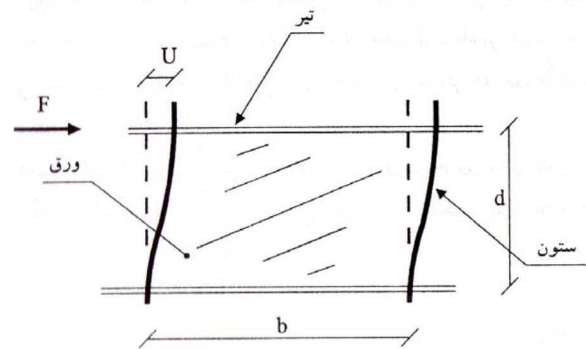
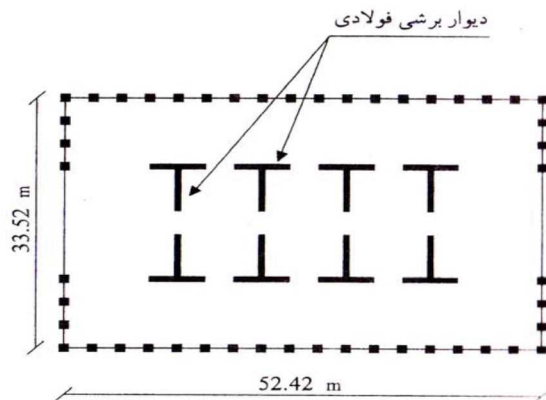


ارتفاع ساختمان	فرم سازه ای مناسب
ساختمانهای تا حدود ۳۵ طبقه	سیستم دیوار برشی
ساختمانهای تا حدود ۶۰ طبقه	سیستم دیوار برشی - قاب
ساختمانهای تا حدود ۷۰ طبقه	سیستم دیوار برشی با بازو
ساختمانهای تا حدود ۱۰۰ طبقه	سیستم دیوار برشی محیطی

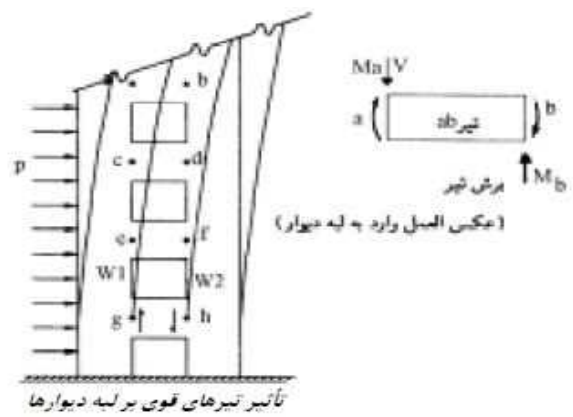




c نمای پشت جزئیات B



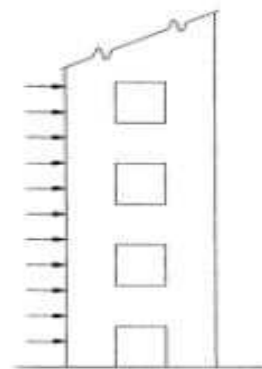
تغییر شکل دیوار برشی کوبیده



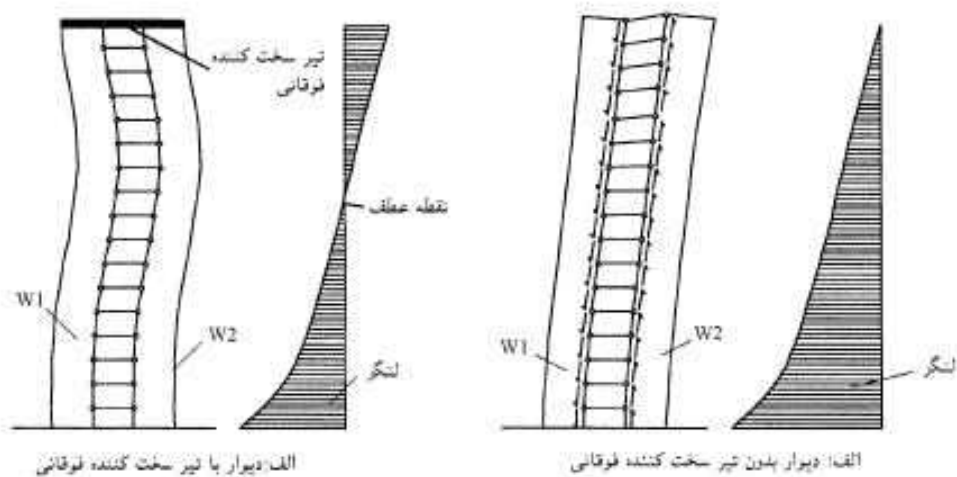
تأثیر تیرهای قوی بر لبه دیوارها



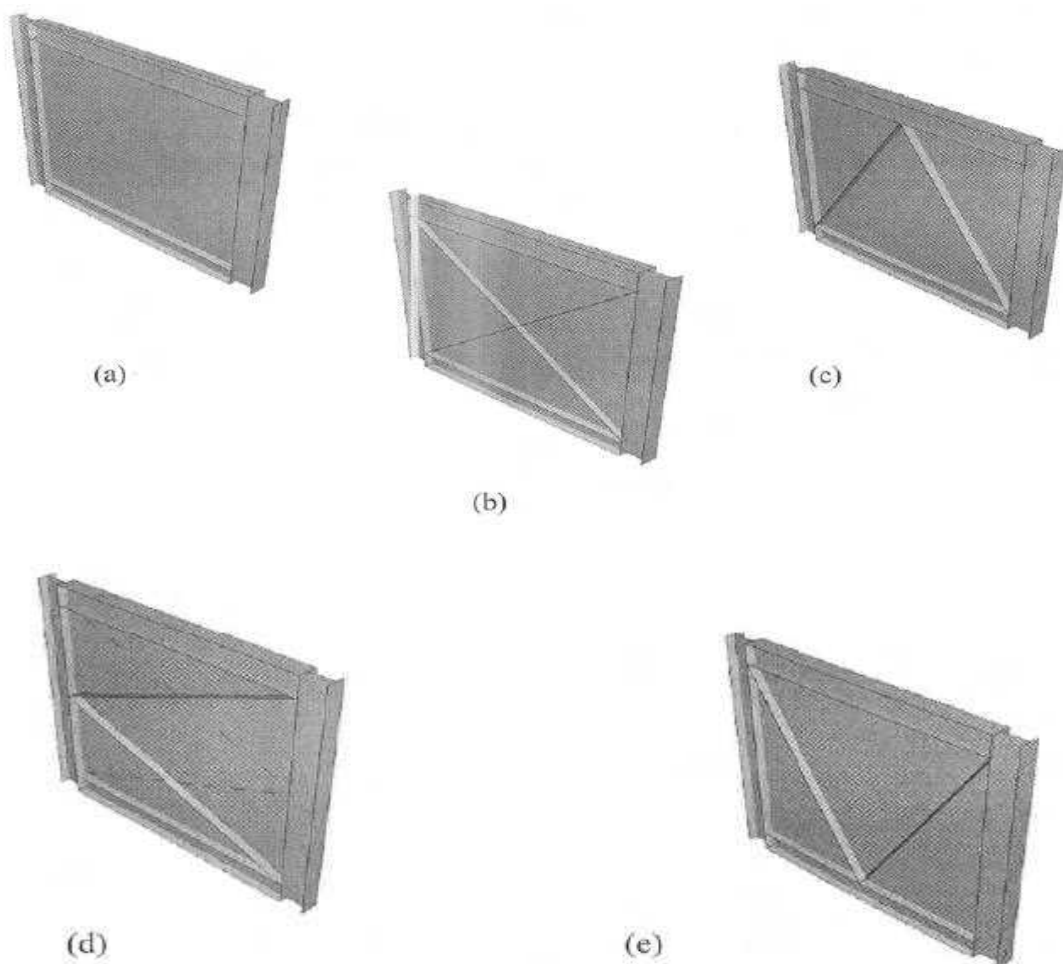
به : تغییر شکل دیوار تحت بارگذاری جانبی



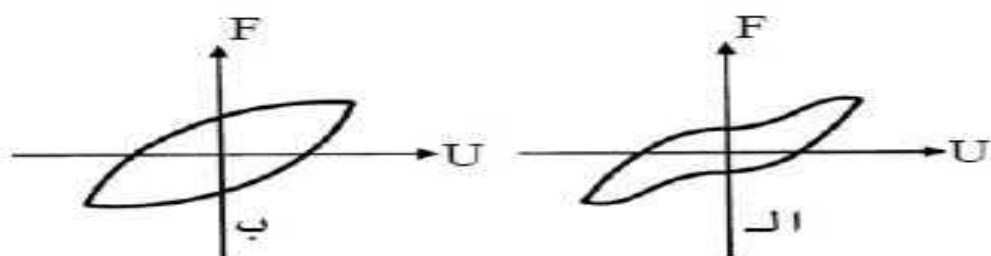
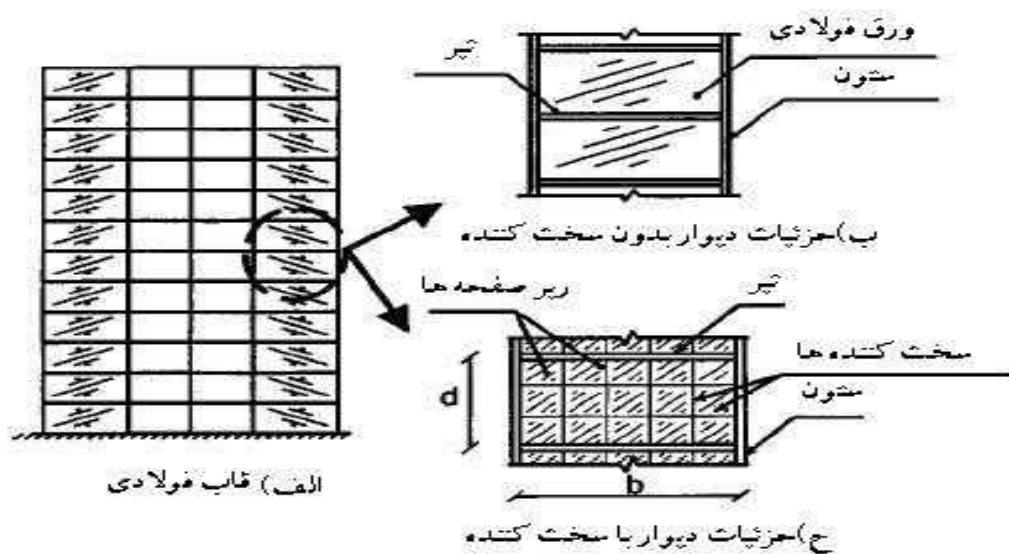
الف : بارگذاری جانبی روی دیوار



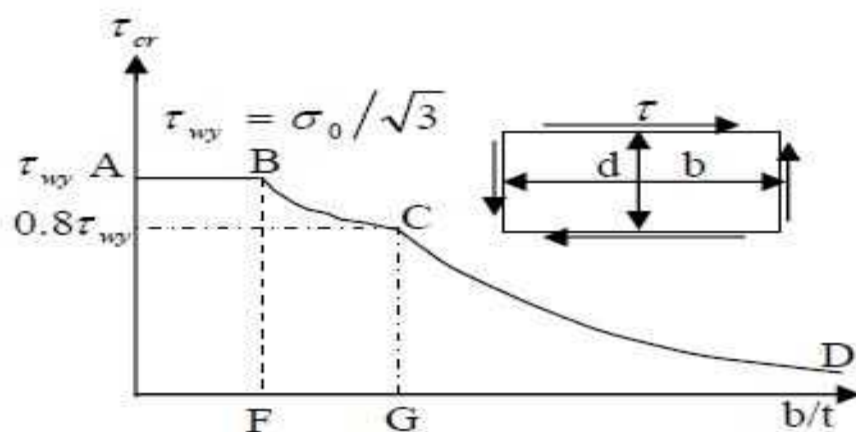
عملکرد تیر فوقانی در تغییرات لنگر در ارتفاع



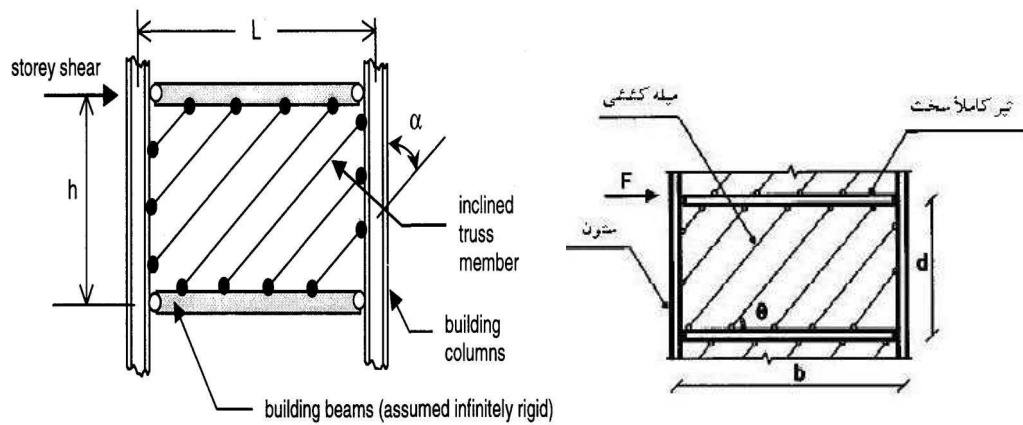
مدل المان محدود SPW : (a) بدون سخت کننده (b) با سخت کننده شکل X (c) با سخت کننده
 V معکوس (d) با سخت کننده شکل K (e) با سخت کننده شکل V



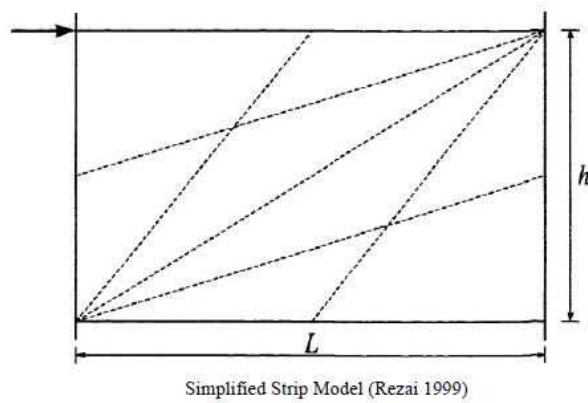
منحنی هیستریزیس دیوار برشی فولادی الف- بدون سخت کننده ب- با سخت کننده



منحنی معیار کماتش برشی در صفحات

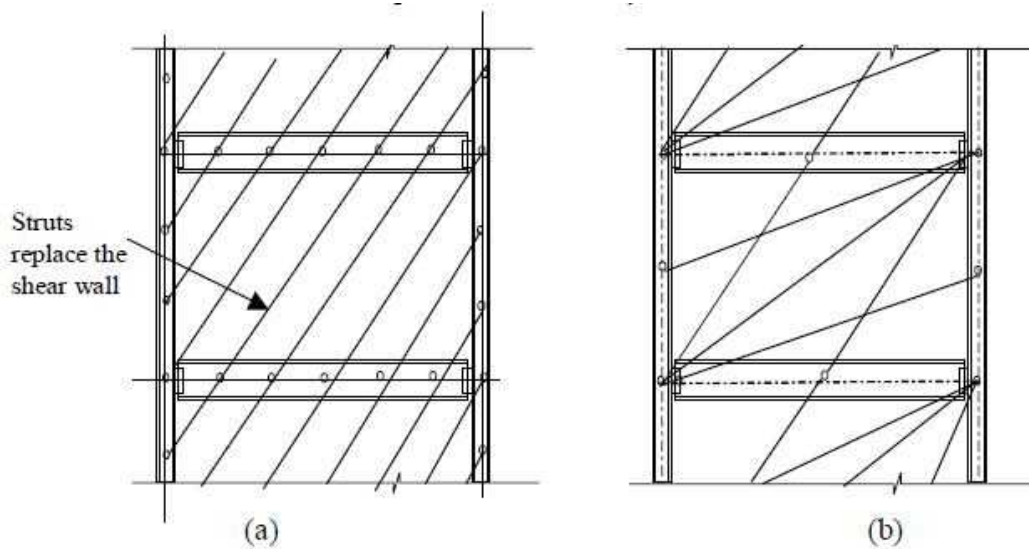


مدل نواری



$$A_s = \frac{(L \cos \alpha + h \sin \alpha) r_w}{n}$$

$$\Delta_x = \frac{1}{n} (L + h \tan \alpha)$$



Two proposed strut models: (a) Thorburn et al. (1983) and (b) Rezaii et al. (2000)

کمانش کلی

$$\tau_{avg} = (D_x^{0.75} D_y^{0.25}) \frac{K_\xi \pi^2}{td^2} \leq \tau_{wy}$$

$$D_x = \frac{EI_x}{s_x} + \frac{Et^3}{12(1-\nu^2)}$$

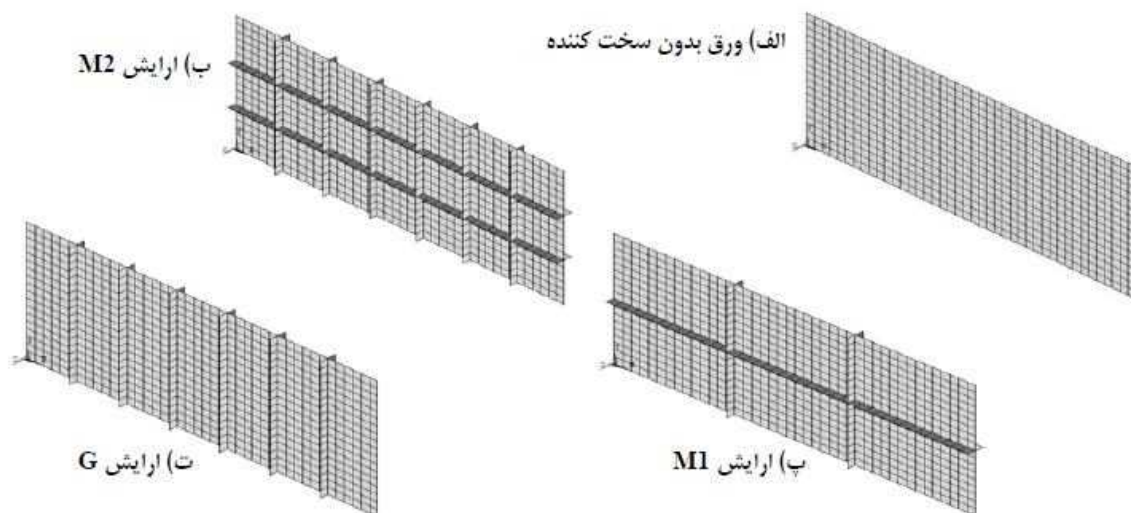
$$D_y = \frac{EI_y}{s_y} + \frac{Et^3}{12(1-\nu^2)}$$

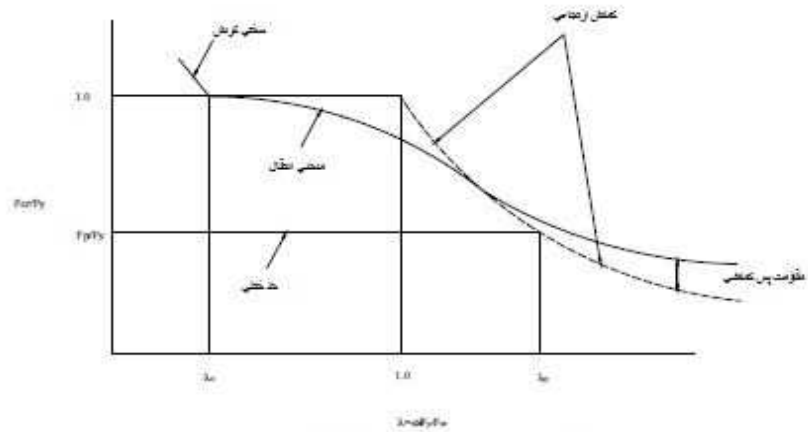
کمانش موضعی

$$\frac{s_y}{s_x} \geq 1 \quad \Rightarrow \quad K_l = 5.35 + 4 \left(\frac{s_x}{s_y} \right)^2$$

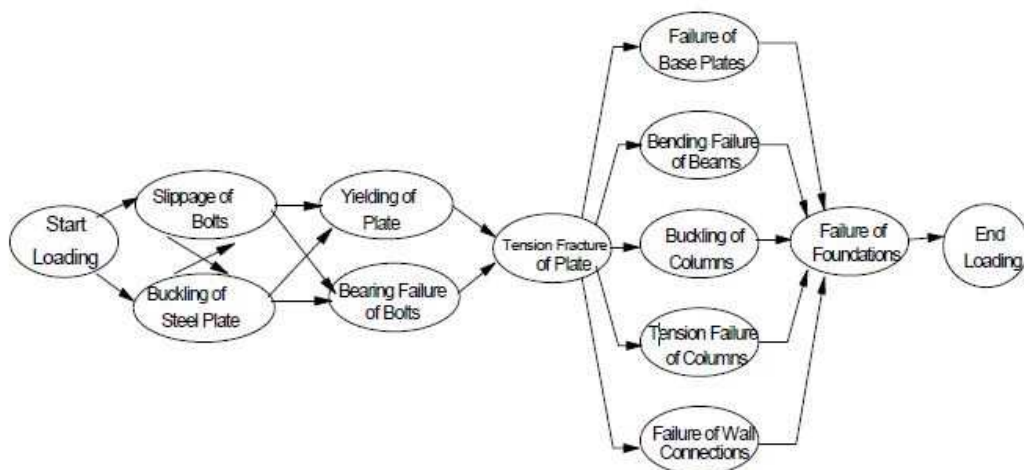
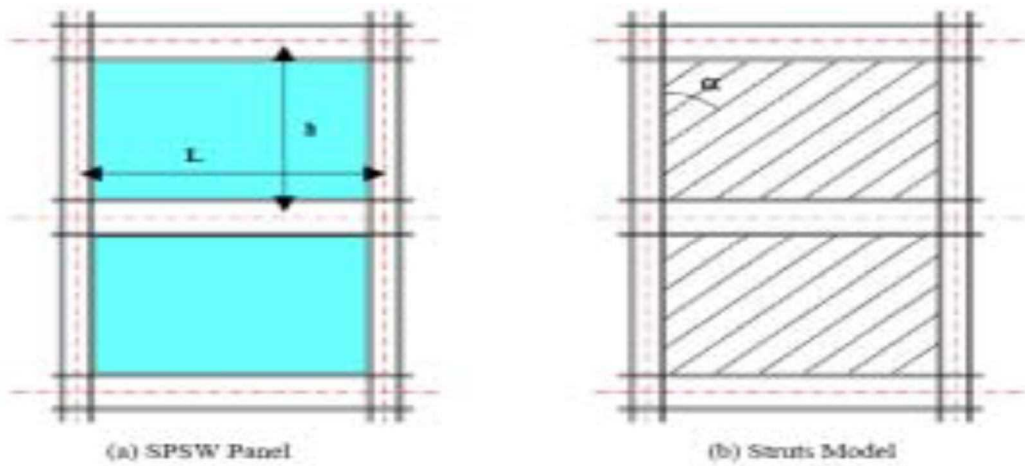
$$\frac{s_y}{s_x} \leq 1 \quad \Rightarrow \quad K_l = 4 + 5.35 \left(\frac{s_x}{s_y} \right)^2$$

$$\tau_{crl} = \frac{K_l \pi^2 E}{12(1-\nu^2)} \left(\frac{t}{s_x} \right)^2 \leq \tau_{sy}$$



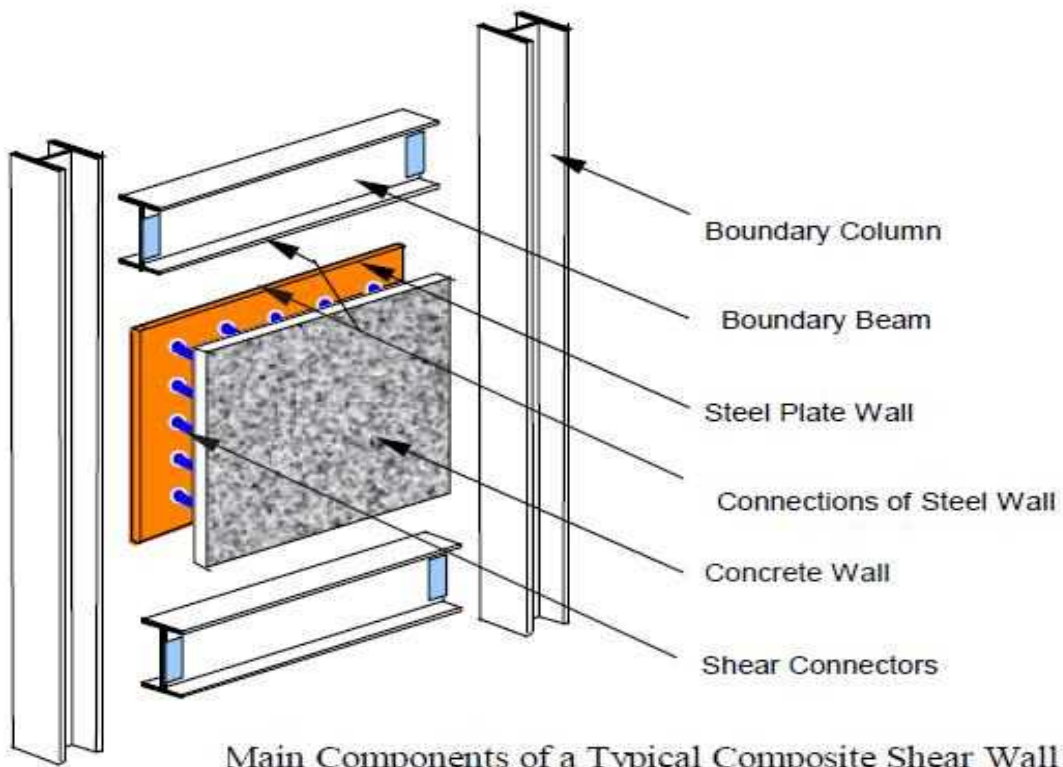
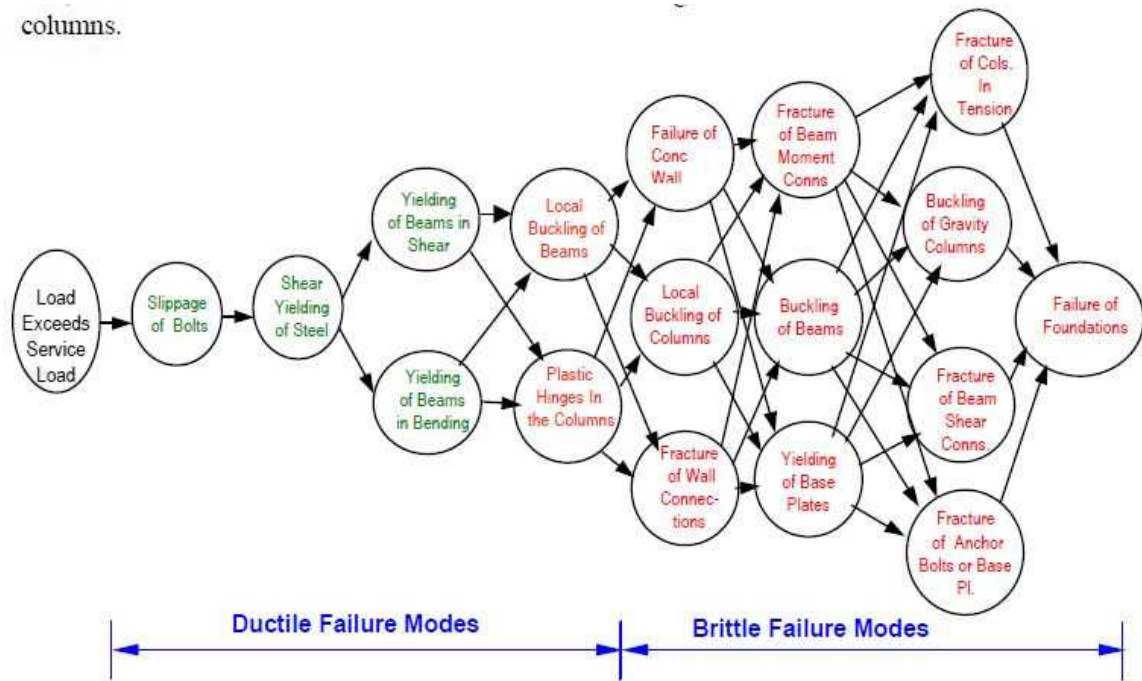


افزایش مقاومت ناشی از اثر پس کشش



Major failure modes of typical steel plate shear walls

columns.



Main Components of a Typical Composite Shear Wall

Samples of steel shear walls with opening

$$\tau_{cr} = \frac{K\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \quad \text{قبل از کمانش}$$

$$K = \begin{cases} 5.35 + 4\left(\frac{b}{d}\right)^2 & \frac{d}{b} \geq 1 \\ 5.35\left(\frac{b}{d}\right)^2 + 4 & \frac{d}{b} \leq 1 \end{cases}$$

$$u_{wcr} = d.\gamma_{cr}, \gamma_{cr} = \frac{\tau_{cr}}{G}, G = \frac{E}{2(1+\mu)}$$

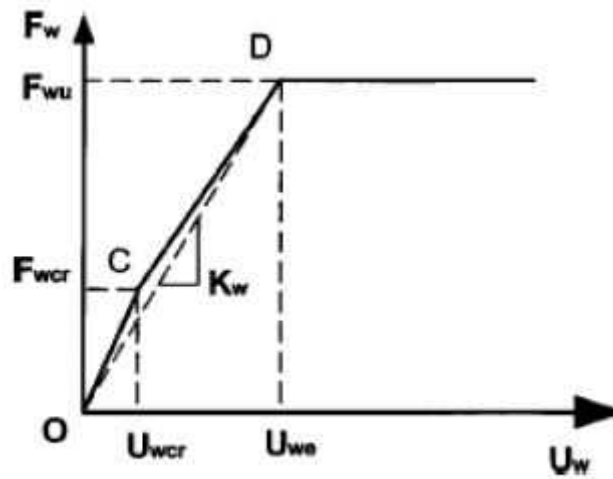
$$U_{wpb} = \frac{2\sigma_{\eta}.d}{E\sin 2\theta}$$

$$U_{we} = \left(\frac{\tau_{cr}}{G} + \frac{2\sigma_{\eta}}{E\sin 2\theta}\right)d$$

► قبل از کمانش ورق : $F_w = \frac{Gbt}{d}U_w$

► پس از کمانش ورق : $F_w = \left(\frac{Ebt}{4d}\right)U_w - \tau_{cr}(bt)\left(1 - \frac{E}{4G}\sin^2 2\theta\right)$

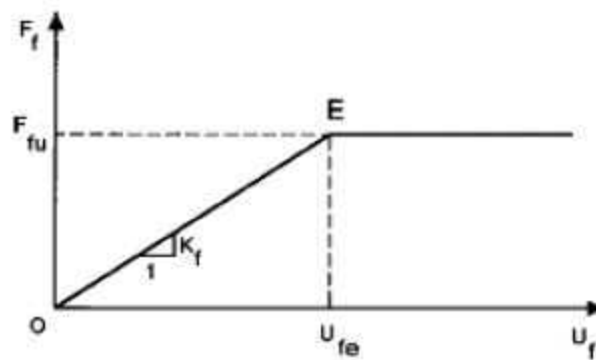
$$K_w = \frac{bt}{d} \frac{\tau_{cr} + \frac{1}{2}\sigma_{\eta}\sin 2\theta}{\frac{\tau_{cr}}{G} + \frac{2\sigma_{\eta}}{E\sin 2\theta}} \quad F_w = K_w U_w$$



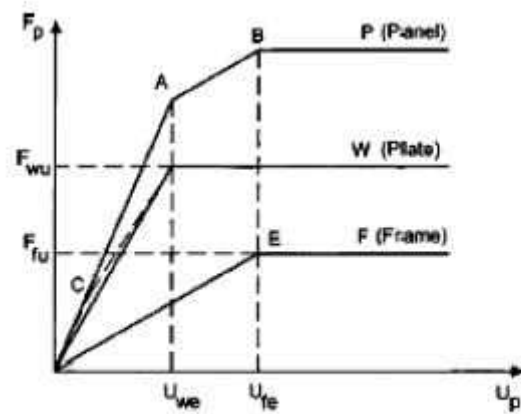
دیاگرام بار- تغییر مکان ورق فولادی

$$F_{fu} = \frac{4M_{fp}}{d} \quad U_{fu} = \frac{M_{fp} \cdot d^2}{6EI_f}$$

$$F_f = K_f U_f \quad k_f = \frac{24EI_f}{d^3}$$



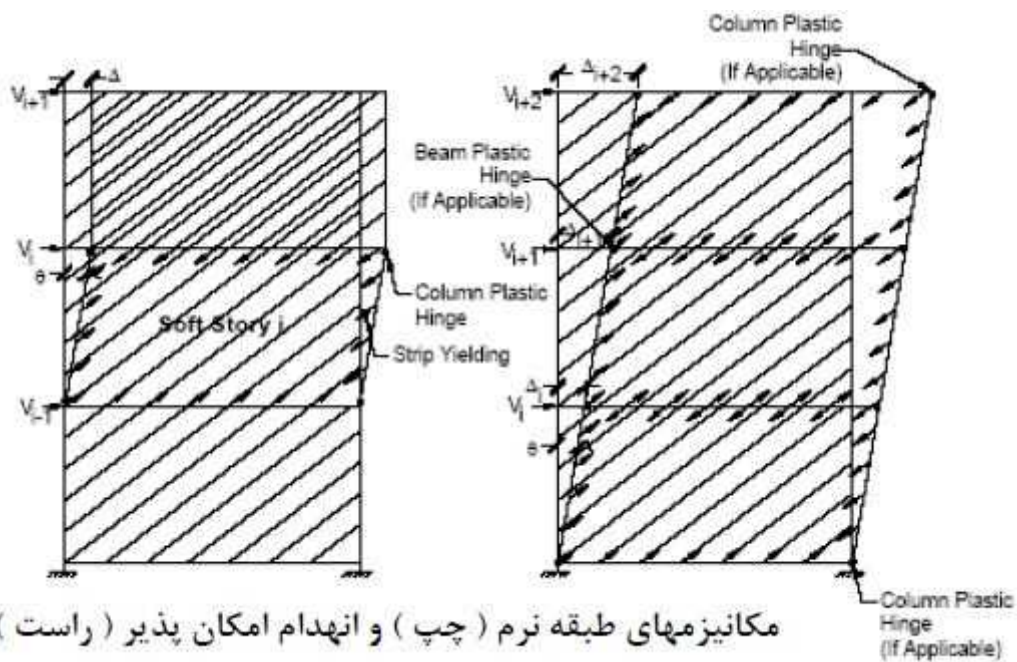
دیاگرام بار- تغییر مکان قاب



دیاگرام مربوط به ورق فولادی ، قاب و مجموع آنها

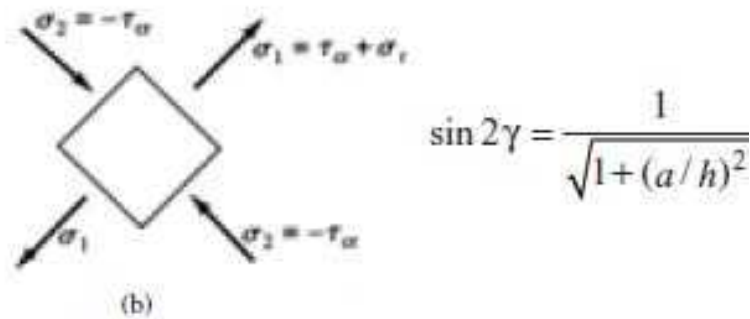
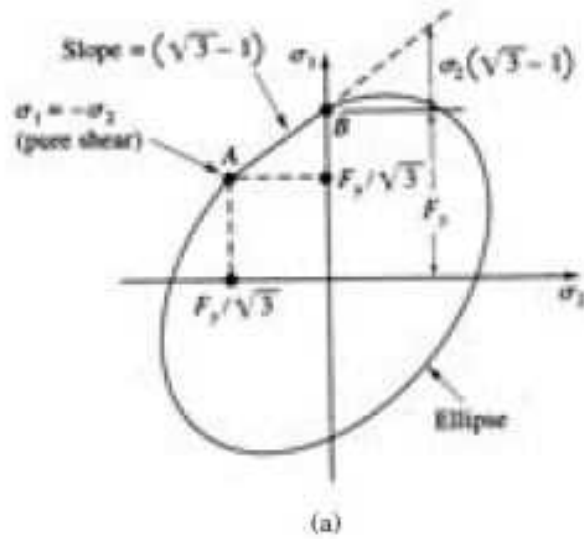
$$V = n_b F_{st} \sin(\alpha) \quad n_b = [L \cos(\alpha)] / s \quad F_{st} = F_y \cdot t \cdot s$$

$$V = \frac{1}{2} F_y t L \sin(2\alpha)$$



مکانیزمهای طبقه نرم (چپ) و انهدام امکان پذیر (راست)

$$t = \frac{2 V_s \Omega_s}{F_y L \sin(2\alpha)}$$



$$V_n = 0.6 F_{yw} A_w \quad \text{for} \quad \frac{h}{t_w} \leq 1.10 \sqrt{\frac{k_v E}{F_{yw}}}$$

$$V_n = 0.6 F_{yw} A_w \left(C_v + \frac{1 - C_v}{1.15 \sqrt{1 + (a/h)^2}} \right) \quad \text{for} \quad \frac{h}{t_w} > 1.10 \sqrt{\frac{k_v E}{F_{yw}}}$$

System Overstrength Factor, Ω_o	
Seismic Load Resisting System	Ω_o
All moment-frame systems meeting Part I requirements	3
Eccentrically Braced Frames (EBF) meeting Part I requirements	$2^{1/2}$
All other systems meeting Part I requirements	2

Failure modes of steel plate wall

1. Slippage of bolts (ductile).
2. Buckling of the steel plate (ductile).
3. Yielding of the steel plate (ductile).
4. Fracture of wall plate (brittle).
5. Fracture of the connections of steel wall to boundary columns and beams (brittle).

Failure modes of top and bottom beams

6. Shear yielding of top and bottom beams (ductile).
7. Plastic hinge formation in top and bottom beams (ductile).
8. Local buckling in the top and bottom beam flanges or web (ductile if $b/t \leq \lambda_p$).
9. Fracture of moment connections of the beams in dual systems (brittle).
10. Overall or lateral-torsional buckling of beams (brittle).
11. Fracture of shear connections of beams (brittle).

Failure modes of Boundary Columns

12. Plastic hinge formation at the top and bottom of columns (ductile).
 13. Local buckling of boundary columns (ductile if $b/t \leq \lambda_p$).
 14. Overall buckling of boundary columns (ductile if $\lambda_c = (KL/\pi r) \sqrt{F_y/E} \leq 1.0$.)
 15. Tension fracture of boundary columns or their splices (brittle).
 16. Yielding of base plates of boundary columns in uplift (ductile)
 17. Fracture of anchor bolts or base plates at the base of columns in uplift (brittle)
 18. Fracture of column base plates in bending and/or uplift (brittle)
 19. Failure of foundations of the wall (brittle).
-
- a. Compressive buckling of the steel plate along the diagonal compression field,
 - b. Slippage of bolts connecting the wall plate to fin-plates on the boundary columns and beams,
 - c. Tension yielding of the steel plate along the diagonal tension field,
 - d. Tension fracture of wall plate along the diagonal tension field,
 - e. Bending failure of boundary beams or their connections,
 - f. Failure of the connections of the wall to boundary columns and beams,
 - g. Compression failure of boundary columns,
 - h. Tension fracture of the boundary columns,,
 - i. Failure of base plates of boundary columns in compression or in uplift; and,
 - j. Failure of foundations of the wall.

حالت های گسیختگی دیوار صفحه ای فولادی

۱- لغزش پیچها (شکل پذیر)

۲- کمپانچ صفحه فولادی (شکل پذیر)

۳- تسلیم شدگی صفحه فولادی (شکل پذیر)

۴- شکست صفحه دیوار (ترد)

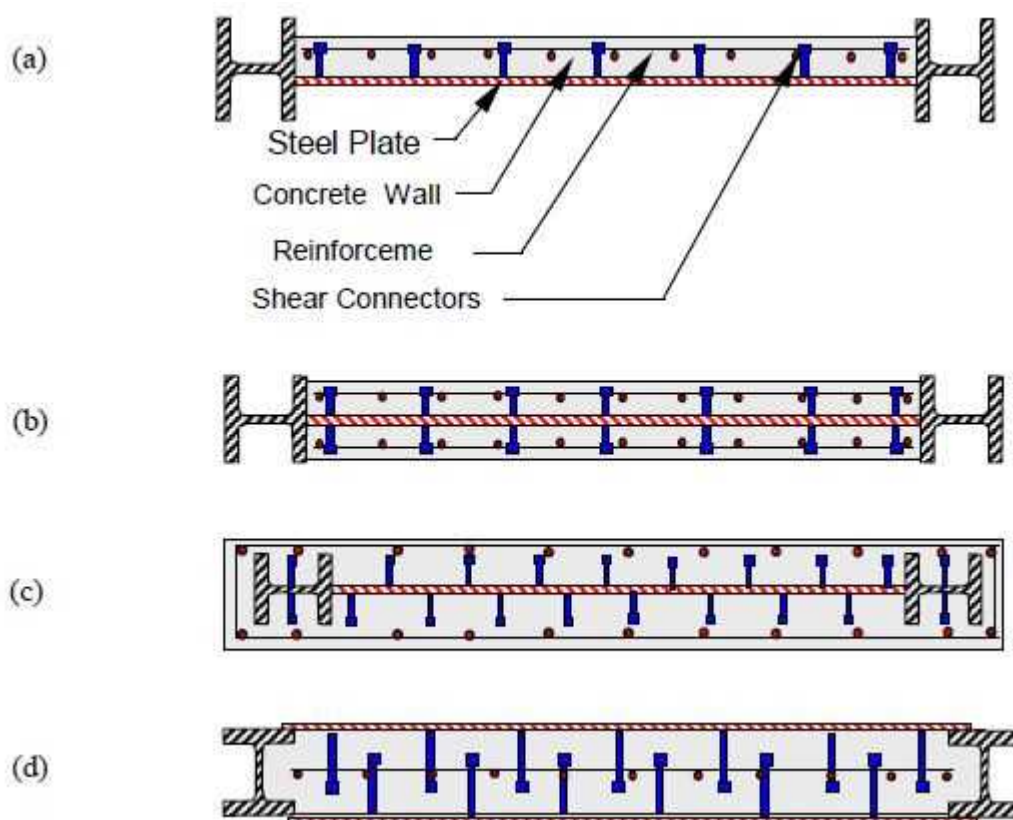
۵- شکست اتصالات دیوار فولادی با تیرها و ستونهای مرزی (ترد)

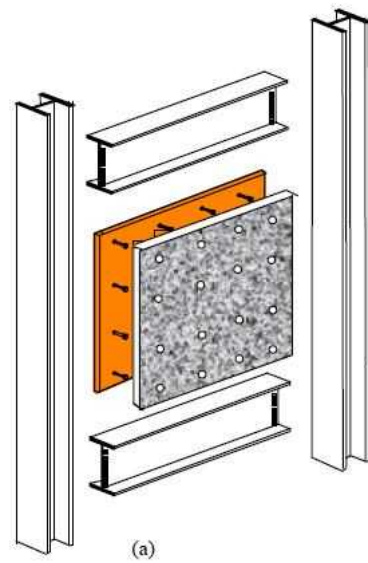
حالتهای گسیختگی تیرهای بالا و پایین

- ۱- تسلیم شدگی برشی تیرهای بالا و پایین (شکل پذیر)
- ۲- تغییر شکل مفصل پلاستیک در تیرهای بالا و پایین (شکل پذیر)
- ۳- کمانش موضعی در بالها یا جان تیرهای بالا و پایین
- ۴- شکست اتصالات خمشی تیرها در سیستمهای دوگانه (ترد)
- ۵- کمانش پیچشی کلی یا جانبی تیرها (ترد)
- ۶- شکست اتصالات برشی تیرها (ترد)

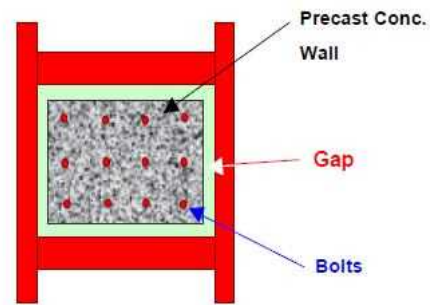
حالتهای گسیختگی ستونهای مرزی

- ۱- تغییر شکل پلاستیک در بالا و پایین ستونها (شکل پذیر)
- ۲- کمانش موضعی ستونهای مرزی
- ۳- کمانش کلی ستونهای مرزی
- ۴- شکست کششی ستونهای کممرزی یا وصله های آنها (ترد)
- ۵- تسلیم شدگی صفحات کف ستونهای مرزی در اثر نیروهای بلند کننده (شکل پذیر)
- ۶- شکست پیچهای مهار یا صفحات کف در کف ستونها در اثر نیروهای بلند کننده (ترد)
- ۷- شکست صفحات کف ستون در خمش و یا اثر نیروهای بلند کننده (ترد)
- ۸- گسیختگی پی های دیوار (ترد)

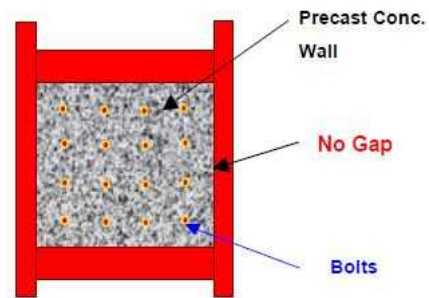




(a) Composite Shear Wall Studied



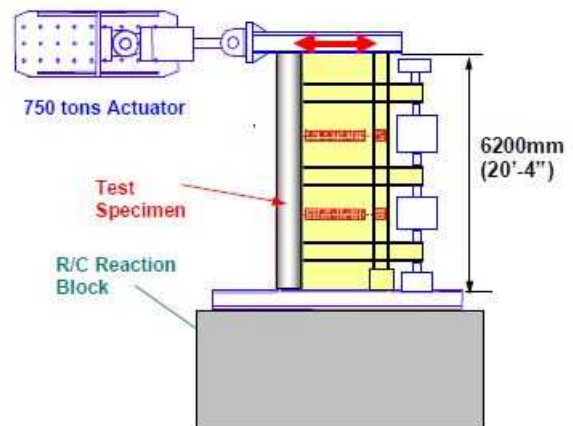
(b) Innovative Composite Wall



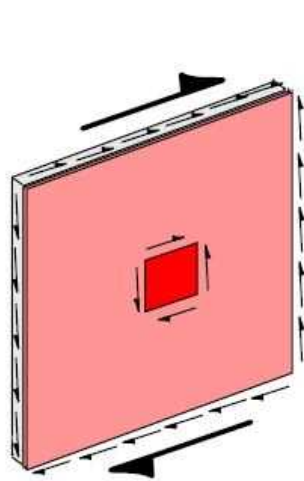
(c) Traditional Composite Wall

Note: Steel shear wall is fillet-welded to steel tab plates on all four boundaries. The tab plates are fillet-welded to the boundary beam and column flanges.

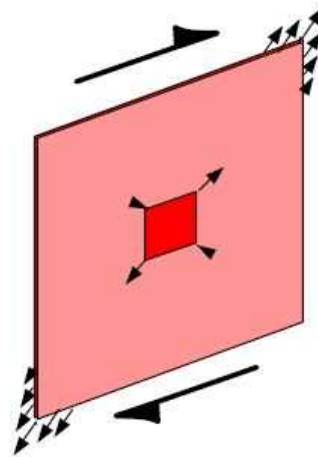
Views of Traditional and "Innovative" Composite Shear Walls



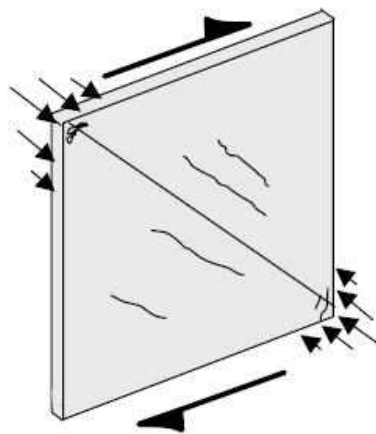
Typical specimen and test set-up (Astaneh-Asl and Zhao, 2000).



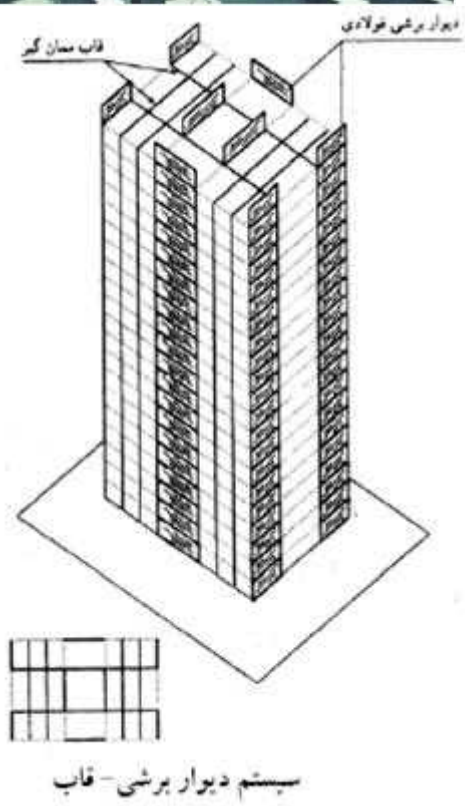
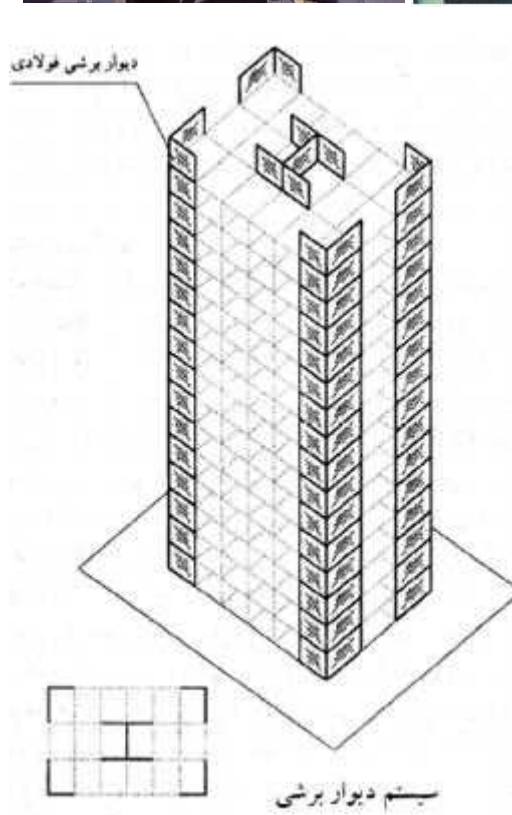
a. Shear Wall Elements Under Pure Shear

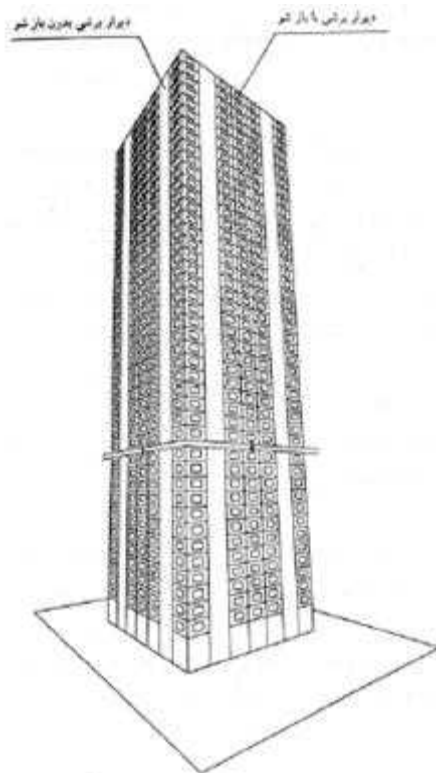


b. Shear Wall Elements Under Tension Field Action

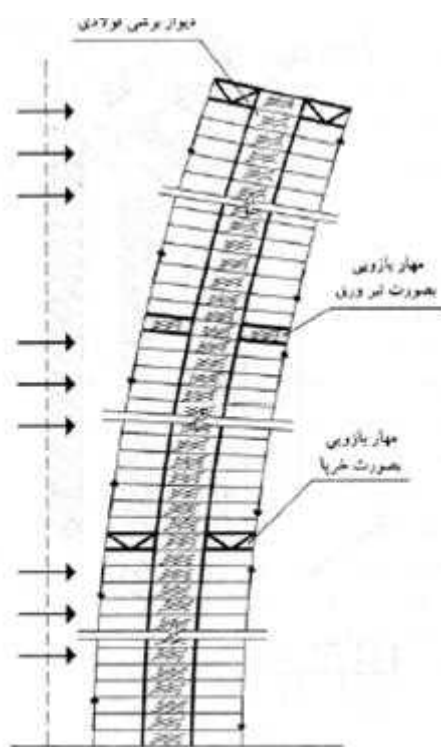


Shear Resisted by Diagonal Compression Field of Concrete

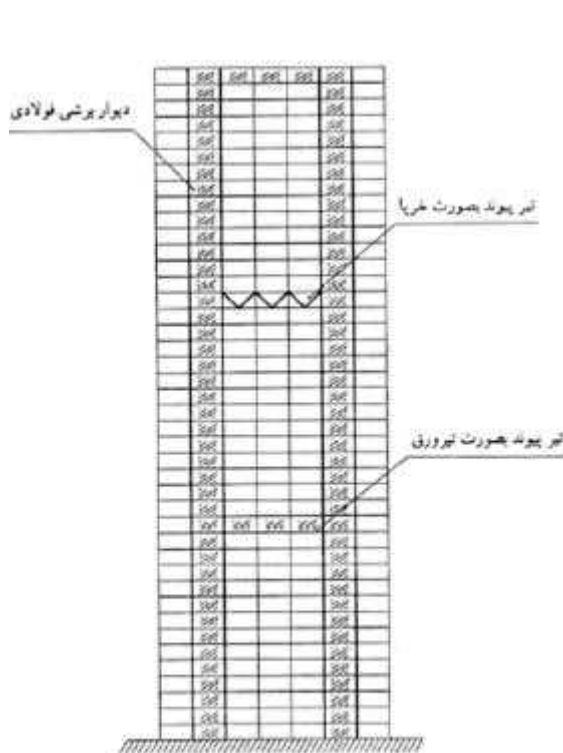




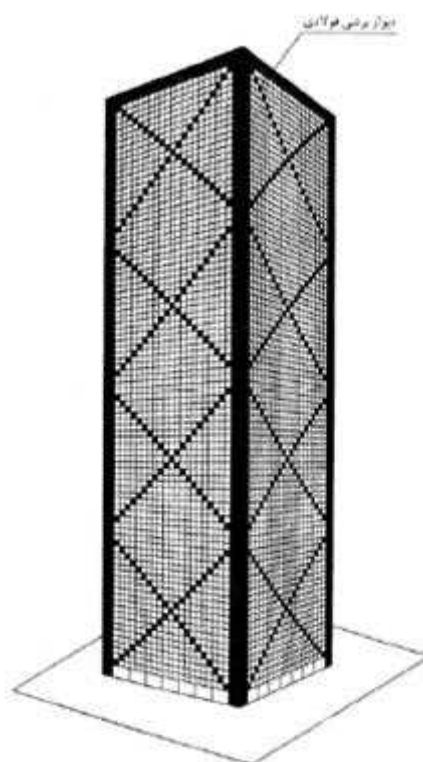
سیستم دیوار برشی محبلی



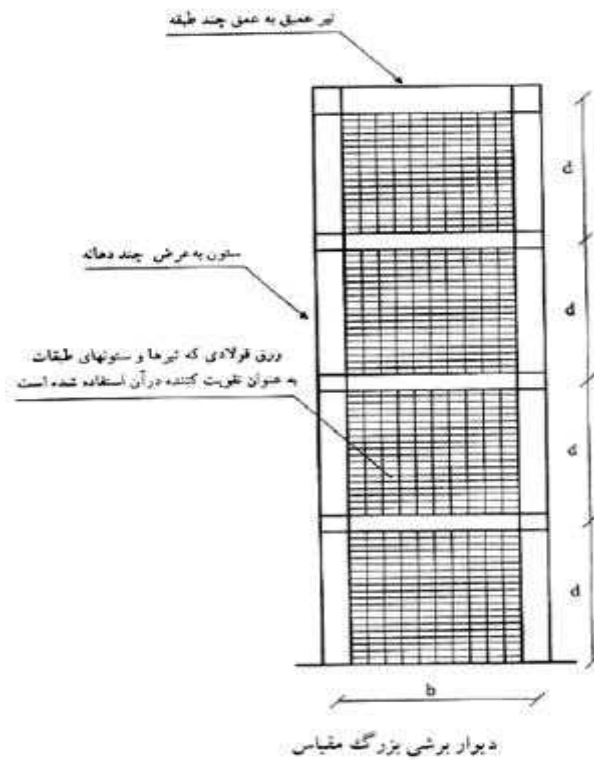
سیستم دیوار برشی با بازو



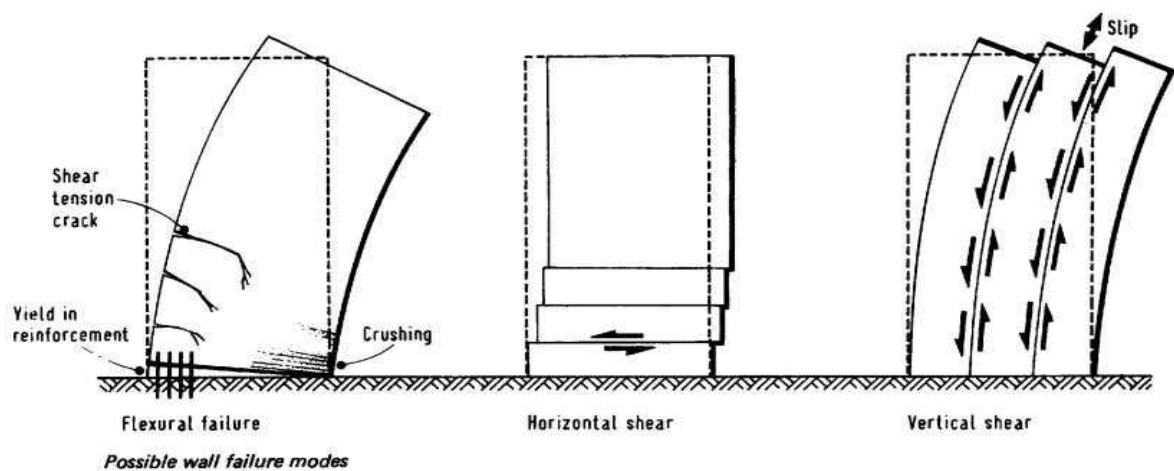
سیستم دیوار برشی بهم پیوسته

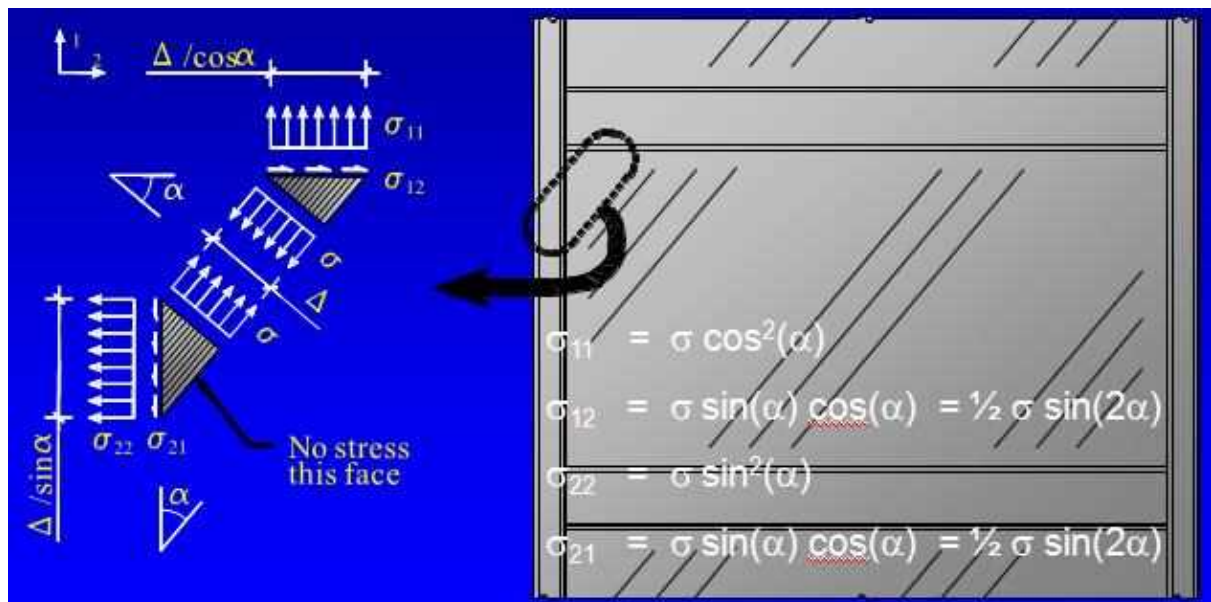
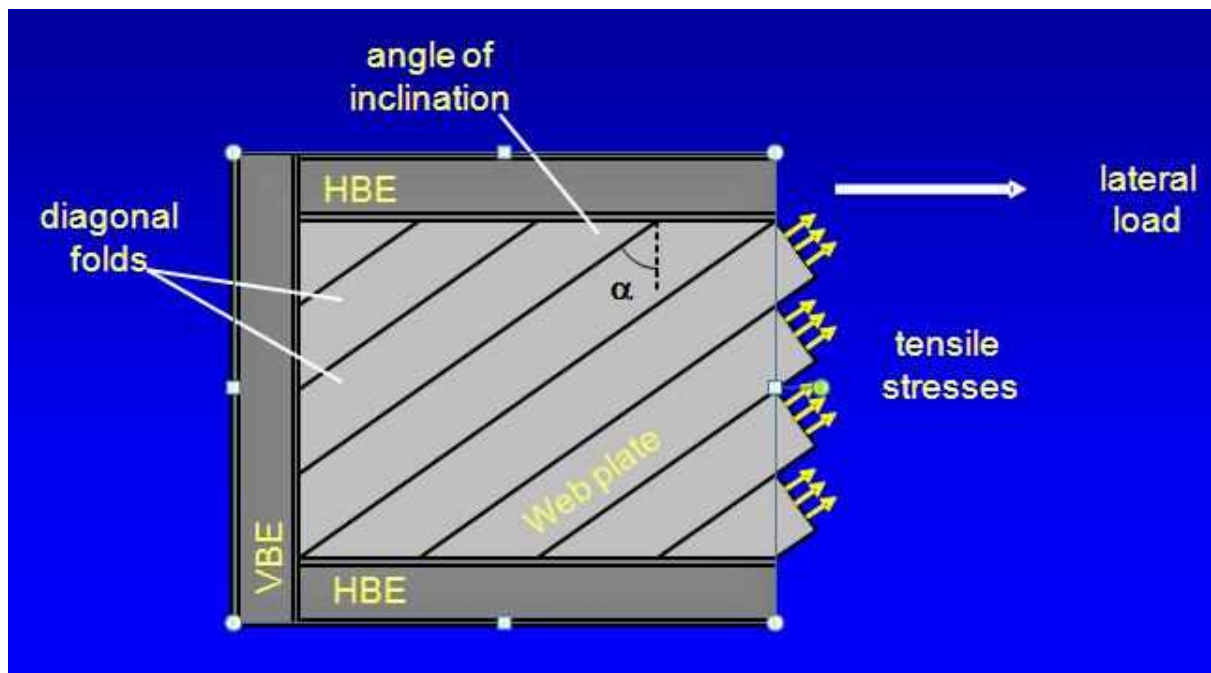


قاب های محبلی پر شده با ورق فولادی

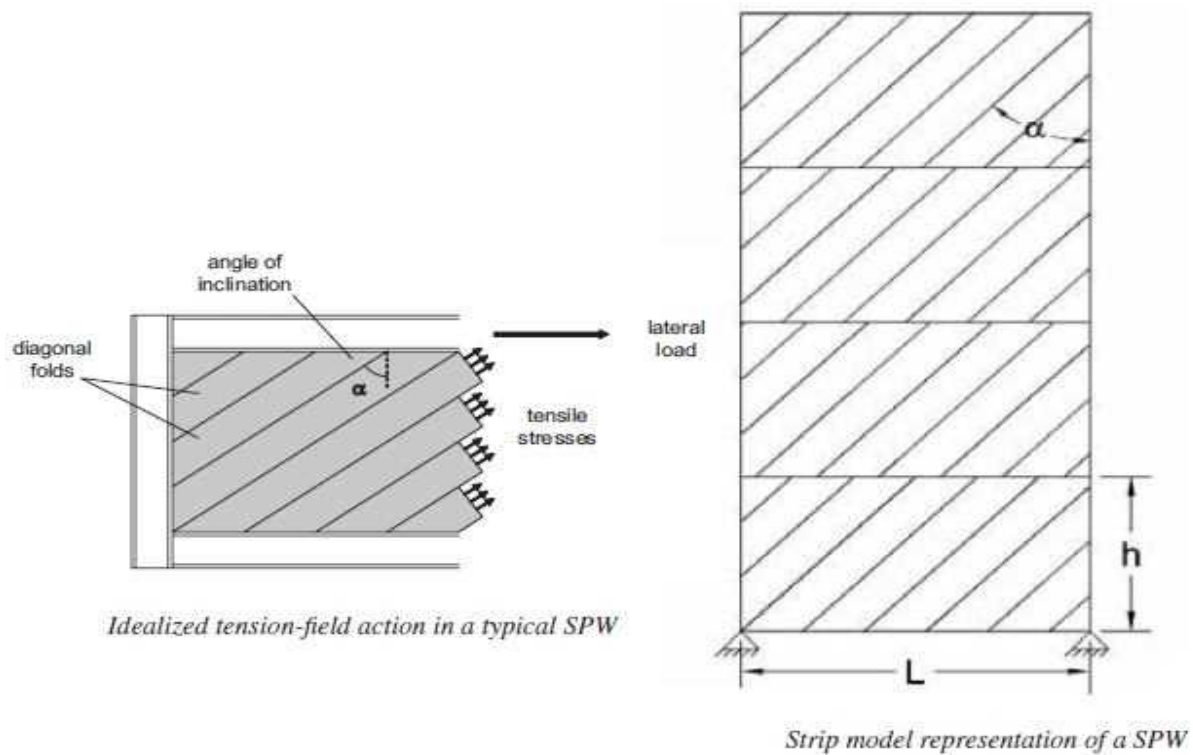


ارتفاع ساختمان	فرم سازه ای مناسب
ساختمانهای تا حدود ۳۵ طبقه	سیستم دیوار برشی
ساختمانهای تا حدود ۶۰ طبقه	سیستم دیوار برشی-قاب
ساختمانهای تا حدود ۷۰ طبقه	سیستم دیوار برشی با بازو
ساختمانهای تا حدود ۱۰۰ طبقه	سیستم دیوار برشی محیطی





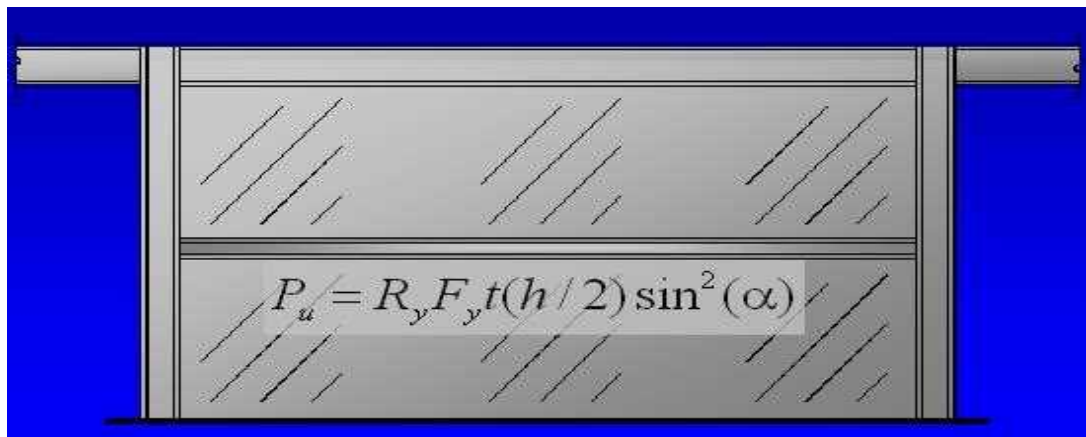
$$\tan^4 \alpha = \frac{1 + \frac{t_w L}{2A_c}}{1 + t_w h \left[\frac{1}{A_b} + \frac{h^3}{360I_c L} \right]}$$



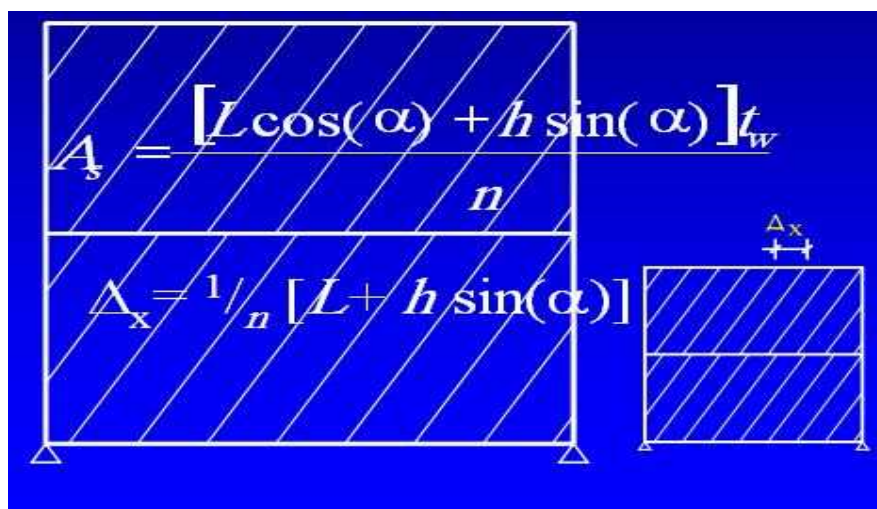
$$w_u = R_y F_y (t_i - t_{i+1}) \cos^2(\alpha)$$

$$I_c \geq 0.00307 \frac{t_w h^4}{L}$$

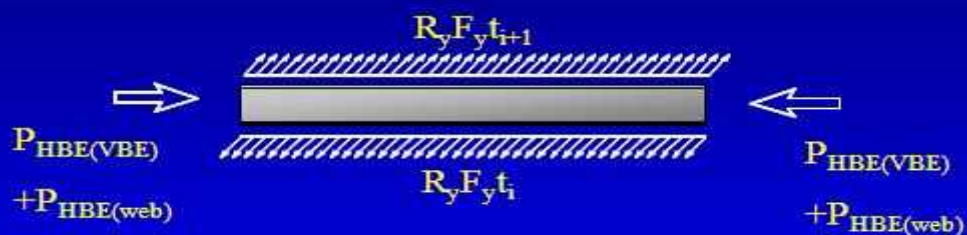
Horizontal Strut



Tension-Strip Model



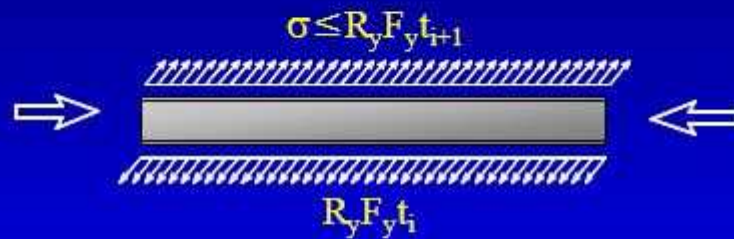
HBE Design Axial Forces



$$P_{HBE(web)} = \frac{1}{2} R_y F_y [t_i \sin(2\alpha_i) - t_{i+1} \sin(2\alpha_{i+1})] L_{cf}$$

$$P_{HBE(VBE)} = \sum \frac{1}{2} h_c R_y F_y \sin^2(\alpha) t_w$$

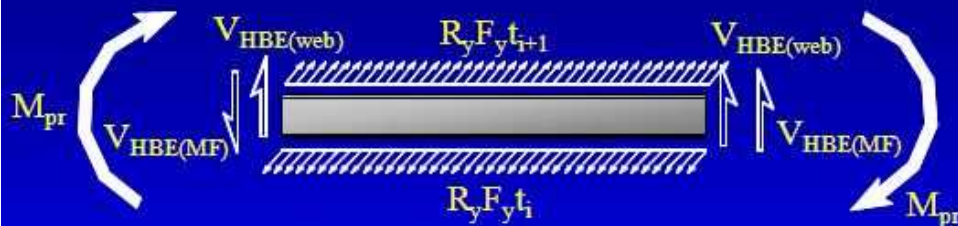
HBE to VBE Design Axial Forces



$$R_{u(horiz)} \geq P_{HBE(VBE)} + \Omega_o P_{collector}$$

$$R_{u(horiz)} \geq P_{HBE(VBE)} + P_{HBE(web)}$$

HBE Design Flexural Forces

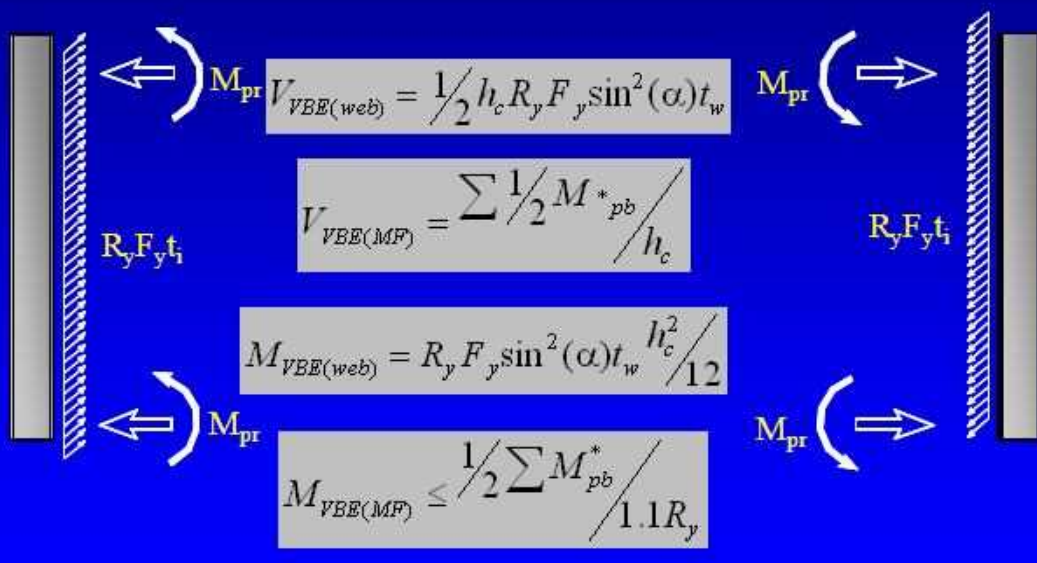


$$M_u = R_y F_y \left[t_i \cos^2(\alpha_i) - t_{i+1} \cos^2(\alpha_{i+1}) \right] L_h^2 / 8 \quad (\text{at midspan})$$

$$V_{HBE(web)} = R_y F_y \left[t_i \cos^2(\alpha_i) - t_{i+1} \cos^2(\alpha_{i+1}) \right] L_h / 2$$

$$V_{HBE(MF)} = 2M_{pr} / L_h$$

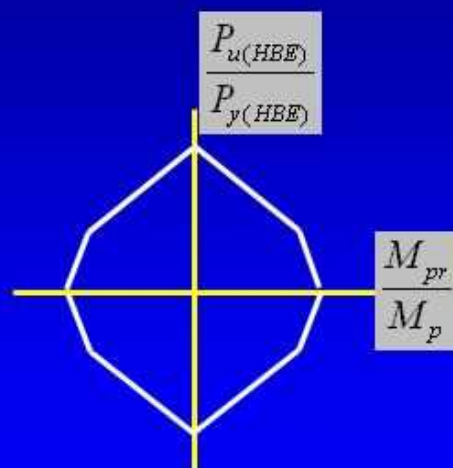
VBE Design Flexural Forces



Flexural strength in the presence of high axial force.

$$M_{pr} \leq [1.1 R_y F_y Z] \left[1 - \frac{1}{2} \frac{P_{u(HBE)}}{P_{y(HBE)}} \right]$$

$$M_{pr} \leq \frac{9}{8} [1.1 R_y F_y Z] \left[1 - \frac{P_{u(HBE)}}{P_{y(HBE)}} \right]$$



Flanges

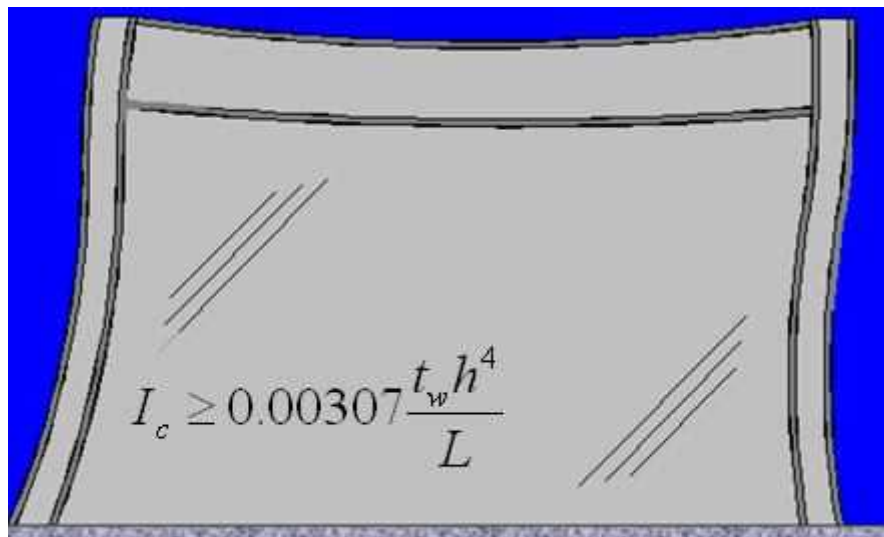
$$\frac{b_f}{2t_f} \leq 0.30 \sqrt{\frac{E_s}{F_y}}$$

Web

$$\frac{P_u}{\phi P_y} \leq 0.125 \quad \frac{h}{t_w} \leq 3.14 \sqrt{\frac{E_s}{F_y}} \left[1 - 1.54 \frac{P_u}{\phi P_y} \right]$$

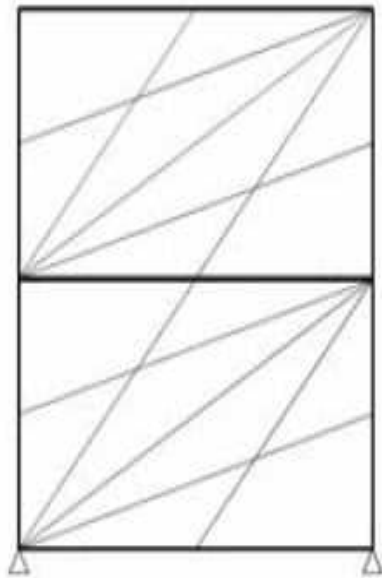
$$\frac{P_u}{\phi P_y} > 0.125 \quad \frac{h}{t_w} \leq 1.12 \sqrt{\frac{E_s}{F_y}} \left[2.33 - \frac{P_u}{\phi P_y} \right] > 1.49 \sqrt{\frac{E_s}{F_y}}$$

$$R_v = 0.6F_y d_c t_p \left[1 + \frac{3b_{cf} t_{cf}^2}{d_b d_c t_p} \right] \left[1.9 - \frac{1.2P_u}{P_y} \right]$$



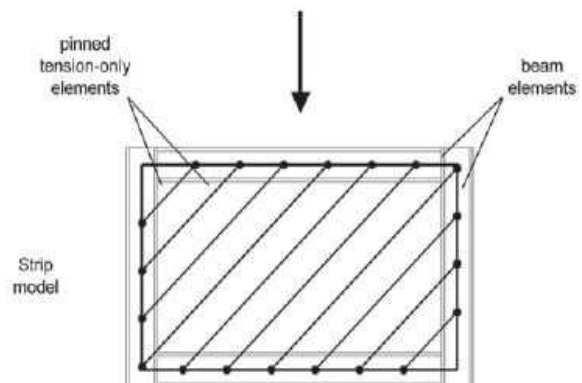
$$\frac{V_{yp \text{ perf}}}{V_{yp}} = \frac{K_{\text{perf}}}{K_{\text{panel}}} = \left[1 - \frac{D}{d} \right]$$

$$\frac{K_{perf}}{K_{panel}} = \frac{1 - \frac{\pi}{4} \left(\frac{D}{S_{diag}} \right)}{1 - \frac{\pi}{4} \left(\frac{D}{S_{diag}} \right) \left(1 - \frac{N_r D \sin \theta}{H_{panel}} \right)} \quad \frac{D}{S_{diag}} \leq \left(1 - Y_t \frac{F_y}{F_u} \right)$$

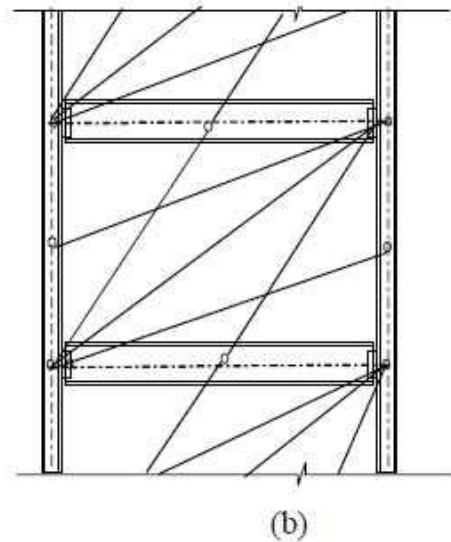
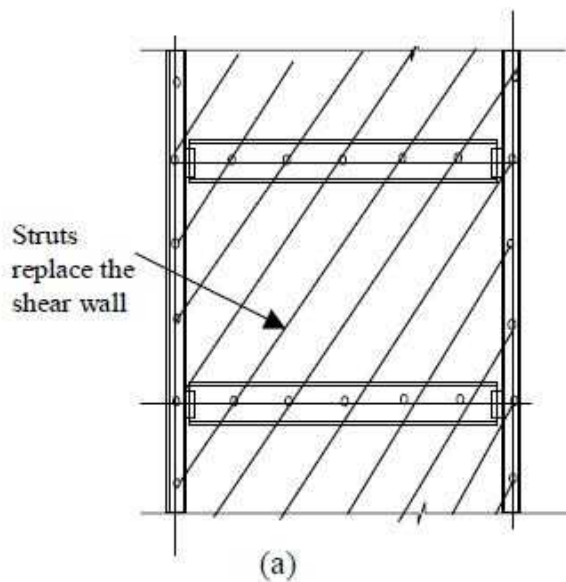


Rezaei's tension-strip model (Rezaei, 1999)

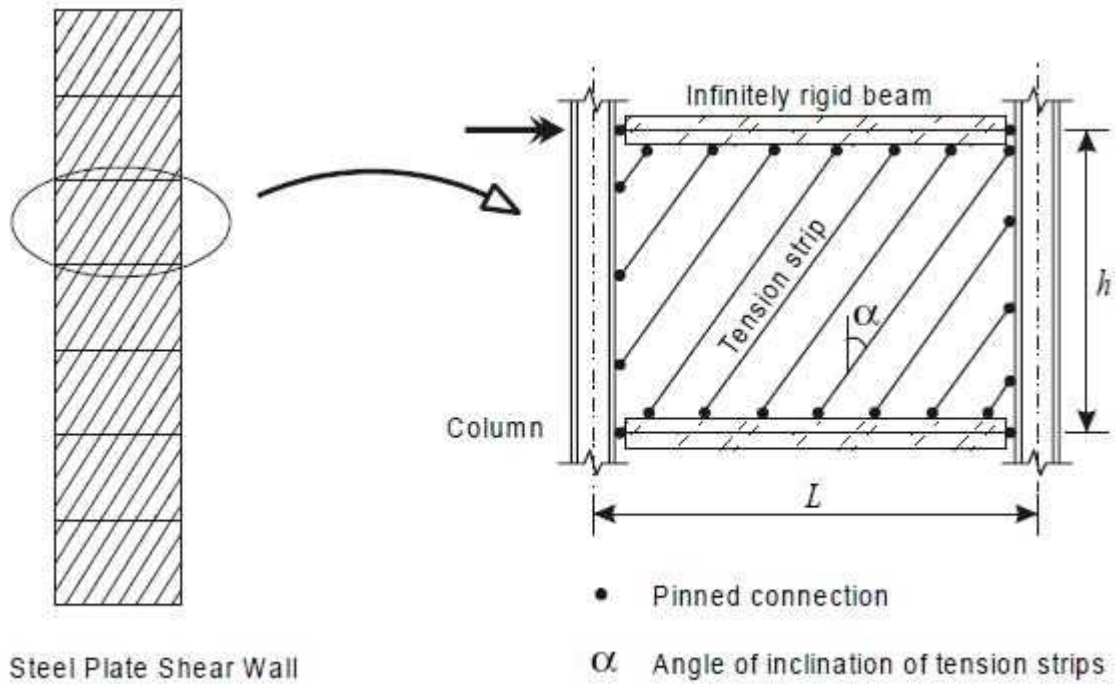
SPSW



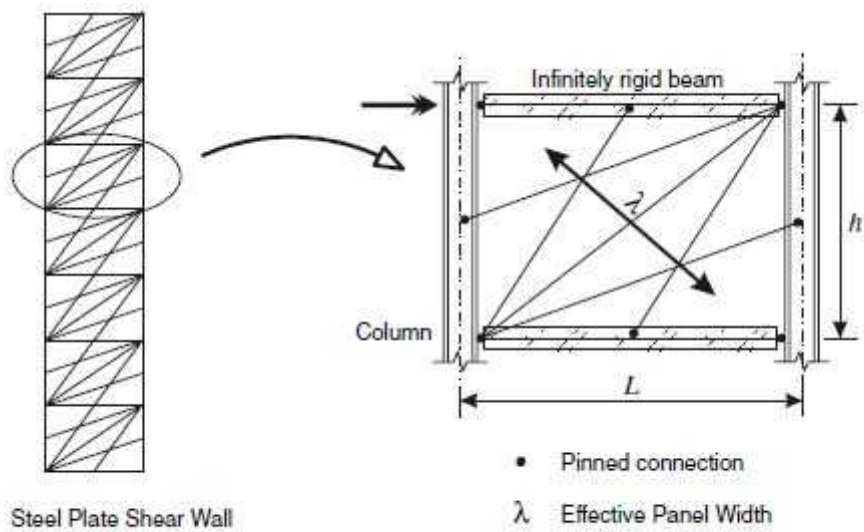
Strip model for static (linear and nonlinear) analysis of SPSW



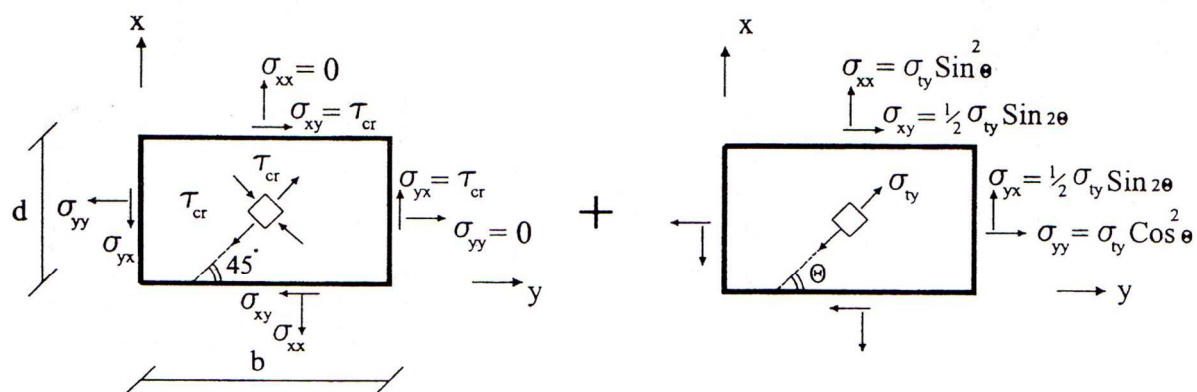
Two proposed strut models: (a) Thorburn et al. (1983) and (b) Rezaei et al. (2000)



Parallel strip model representation of a typical steel plate shear wall panel

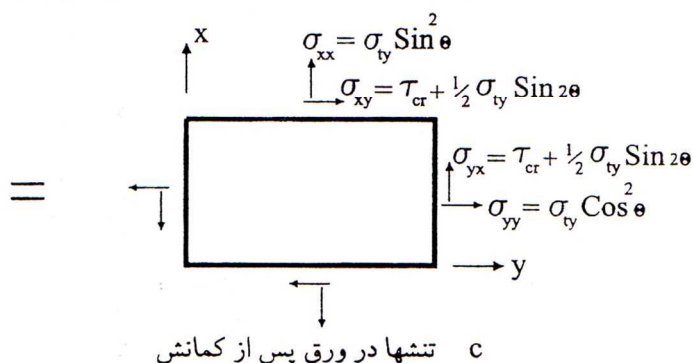


Proposed strip model simulating post-buckling behaviour of a thin steel plate, UBC Model (1999)



a تنشها در ورق هنگام کمانش

b تنش‌های میدان کششی



c تنشها در ورق پس از کمانش

$$\tau_{cr} = \frac{K\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2$$

$$K = 5.35 + 4\left(\frac{b}{d}\right)^2 \quad \text{برای} \quad \frac{d}{b} \geq 1$$

$$K = 5.35\left(\frac{b}{d}\right)^2 + 4 \quad \text{برای} \quad \frac{d}{b} \leq 1$$

$$R = \frac{U_{Max}}{S U_y} \quad F_{wcr} = \tau_{cr} b t$$

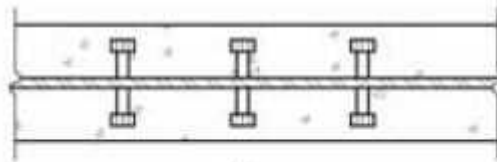
$$V = \frac{1}{2} F_y t L \sin 2\alpha + \frac{4M_P}{h} \quad t_{wi} = \frac{2V_i}{F_y L \sin(2\alpha_i)}$$

$$0.70h \left(\frac{t_w}{2L I_c} \right)^{0.25} \leq 2.5 \quad I_c \geq \frac{0.00307 t_w h^4}{L}$$

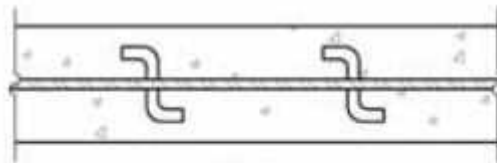
$$\frac{\min(L, h)}{t_w} \leq 25 \sqrt{\frac{E}{F_y}}$$

$$\tan^4 \alpha = \frac{1 + t_w L \left[\frac{1}{2A_c} + \frac{L^3}{120I_b h} \right]}{1 + t_w h \left[\frac{1}{A_b} + \frac{h^3}{360I_c L} \right]} \quad \frac{s}{t_w} \leq 3.82 \sqrt{\frac{E}{F_y}}$$

$$j = 2.5(h/a)^2 - 2 \geq 0.5 \quad I_{st} \geq a t_w^3 j$$



(a)

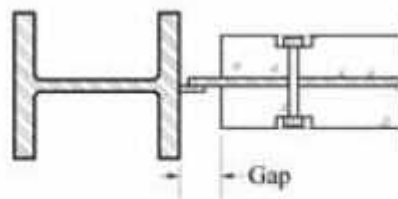


(b)

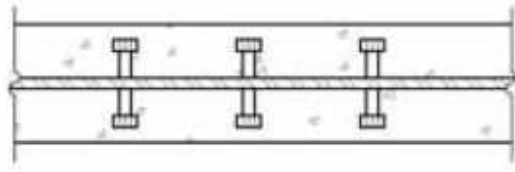


(c)

Types of connectors of concrete to steel web plate in C-SPW.



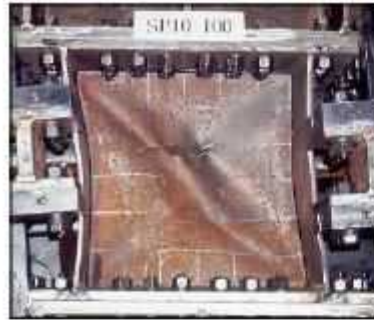
Detail of a C-SPW with a gap between the concrete elements and the boundary elements.



C-SPW with concrete on both sides of the web plate



C-SPW with concrete on one side of the web plate.



(a) SP10-100



(b) SP15-150



(c) SP10-133



(d) SP15-200



(e) SP10-200



(f) SP15-300

$$\tan^4 \alpha = \frac{1 + br \left(\frac{1}{2A_c} + \frac{b^3}{120I_b h} \right)}{1 + hr \left(\frac{1}{2A_b} + \frac{h^3}{320I_c} b \right)}$$

$$\text{For ASD: } t_w \geq \frac{\Omega V_a}{0.42 F_y L_{ef} \sin(2\alpha)}$$

$$I_c \geq 0.00307 \frac{t_w h^4}{L}$$

$$w_{HBE} = \frac{\sigma \cos(\alpha) t_w \sqrt{2}}{\phi 0.6 F_{EXX} [1 + 0.5 \cos^{1.5}(\alpha)]}$$

$$w_{VBE} = \frac{\sigma \sin(\alpha) t_w \sqrt{2}}{\phi 0.6 F_{EXX} [1 + 0.5 \sin^{1.5}(\alpha)]}$$

and for ASD

$$w_{HBE} = \frac{\Omega \sigma \cos(\alpha) t_w \sqrt{2}}{0.6 F_{EXX} [1 + 0.5 \cos^{1.5}(\alpha)]}$$

and for ASD

$$w_{VBE} = \frac{\Omega \sigma \sin(\alpha) t_w \sqrt{2}}{0.6 F_{EXX} [1 + 0.5 \sin^{1.5}(\alpha)]}$$

For LRFD

$$C_a = \frac{P_u}{\phi_b P_y}$$

For ASD

$$C_a = \frac{\Omega_b P_a}{P_y}$$

for $C_a \leq \frac{1}{8}$

$$\frac{h}{t_w} \leq 3.14 \sqrt{\frac{E}{F_y}} [1 - 1.54 C_a]$$

for $C_a > \frac{1}{8}$

$$\frac{h}{t_w} \leq 1.12 \sqrt{\frac{E}{F_y}} [2.33 - C_a]$$

$$t \geq \frac{d_z + w_z}{90} \quad \frac{b_f}{2t_f} \leq 0.30 \sqrt{\frac{E}{F_y}}$$

$$L_b \leq 0.086 r_y \left(\frac{E}{F_y} \right) \quad P_{br} = 0.02 F_y b_f t_f$$

$$\beta_{br} = \frac{13.3 R_y F_y Z}{L_b (d - t_f)} \quad R_u = \frac{1}{2} R_y F_y A_f$$

$$t_w{}_{HBE} \geq \frac{t_w R_y F_y}{F_y{}_{HBE}}$$

$$V_u = V_{VBE(web)} + V_{VBE(HBE)}$$

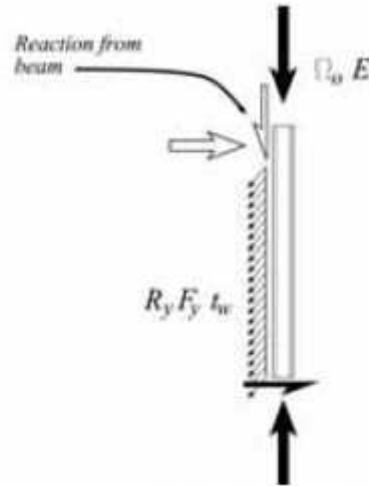
$$V_{VBE(web)} = \frac{1}{2} R_y F_y \sin^2(\alpha) t_w h_c$$

$$V_{VBE(HBE)} = \sum \frac{\frac{1}{2} M_{pb}^{\star}}{h_c} \quad M_{VBE(web)} = \frac{R_y F_y \sin^2(\alpha) t_w h_c^2}{12}$$

$$E_m = \sum \frac{1}{2} R_y F_y \sin(2\alpha) t_w h_c + \sum \left[\frac{2 M_{pr}}{L_h} - \frac{w_u}{2} L_{cf} \right]$$

$$E_m = \frac{1}{2} R_y F_y \sin(2\alpha) t_w h_c + \Omega_0 E_{(above)}$$

$$M_{VBE(HBE)} \leq \frac{\frac{1}{2} \sum M_{pb}^{\star}}{1.1 R_y} \quad R_u = \frac{\sum M_{pb}^{\star}}{d_z} - \frac{1}{2} V_{VBE(HBE)}$$



Free-body-diagram of column under LE+CD loading.

For LRFD

$$Z_c \geq \frac{1}{2} \left[\frac{\sum M_{pb}^*}{2F_{yc} - \frac{|P_{uC}| + |P_{uT}|}{A_g}} \right]$$

For ASD

$$Z_c \geq \frac{1}{2} \left[\frac{\sum M_{pb}^*}{2F_{yc} - \frac{|1.5P_{aC}| + |1.5P_{aT}|}{A_g}} \right]$$

For LRFD

$$s_x \geq \frac{d_h + 1/16 \text{ in.}}{\left[1 - \frac{R_y F_y t_w}{\phi R_t F_u t'_w} \right]}$$

For ASD

$$s_x \geq \frac{d_h + 1/16 \text{ in.}}{\left[1 - \frac{\Omega R_y F_y t_w}{1.5 R_t F_u t'_w} \right]}$$

For LRFD

$$s_x \leq \frac{R_y F_y \cos(\alpha) t_w}{\phi r_n}$$

For ASD

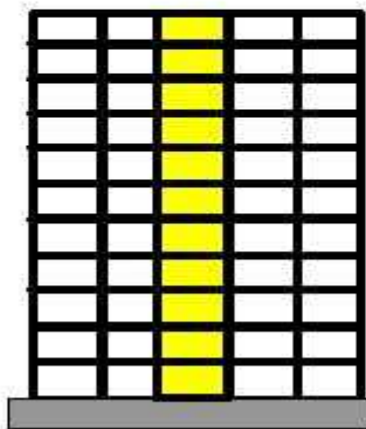
$$s_x \leq \frac{\Omega R_y F_y \cos(\alpha) t_w}{1.5 r_n}$$

For LRFD

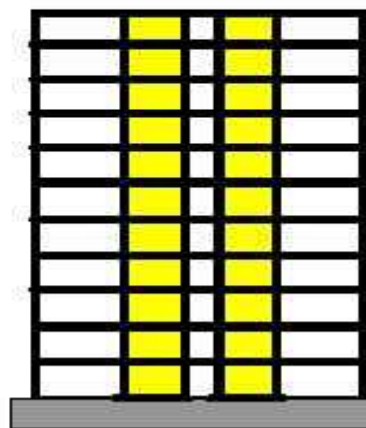
$$s_y \leq \frac{R_y F_y \sin(\alpha) t_w}{\phi r_n}$$

For ASD

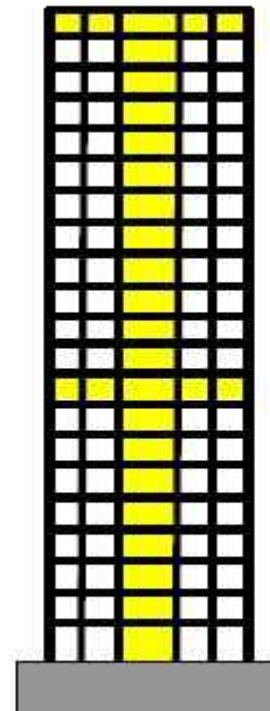
$$s_y \leq \frac{\Omega R_y F_y \sin(\alpha) t_w}{1.5 r_n}$$



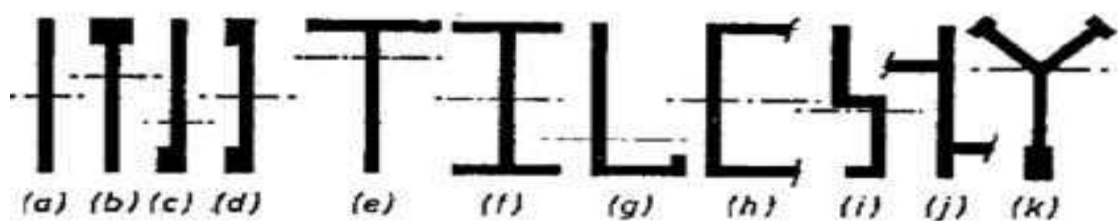
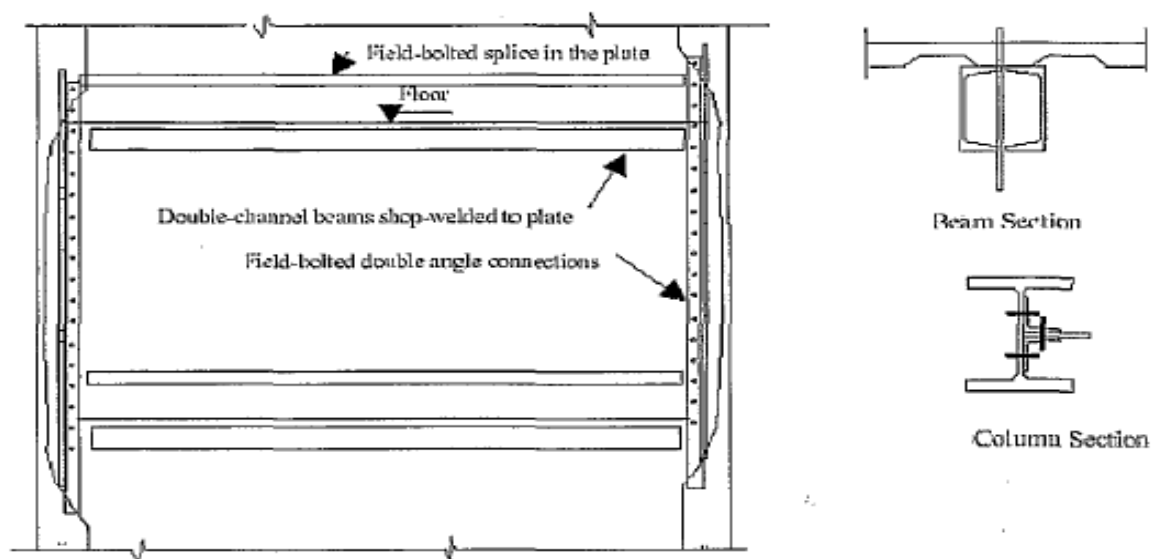
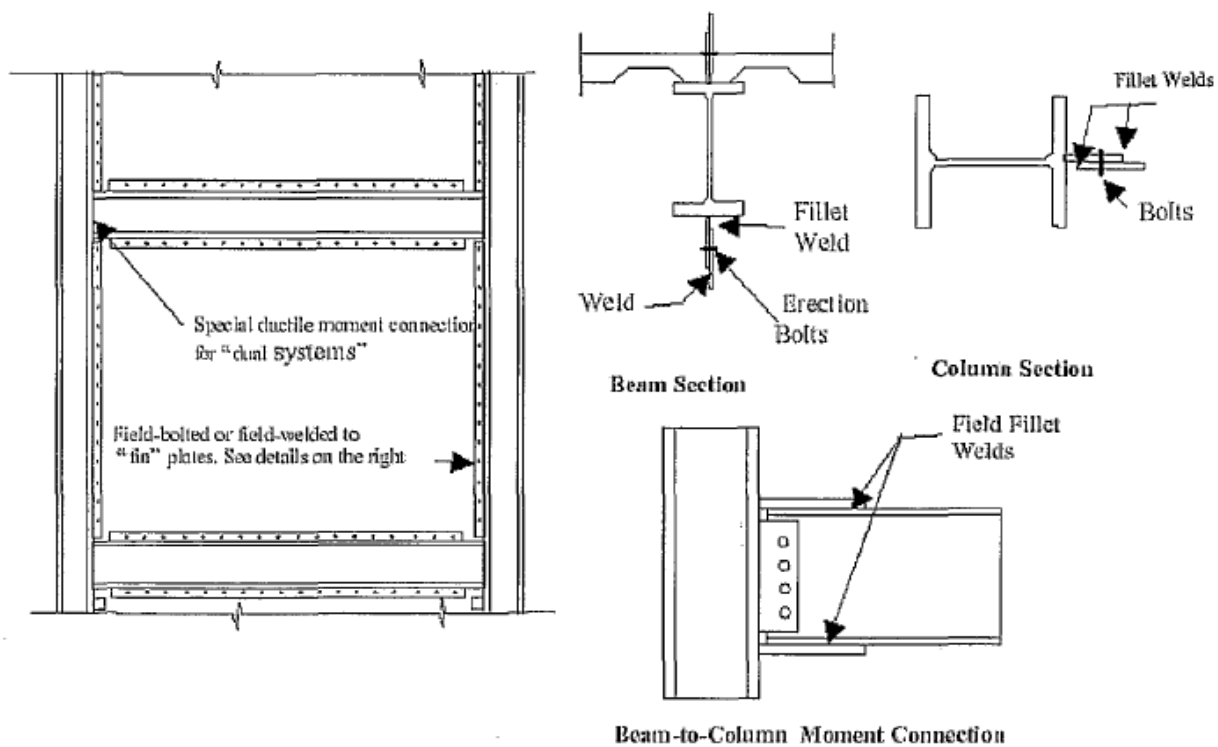
Standard Shear Wall

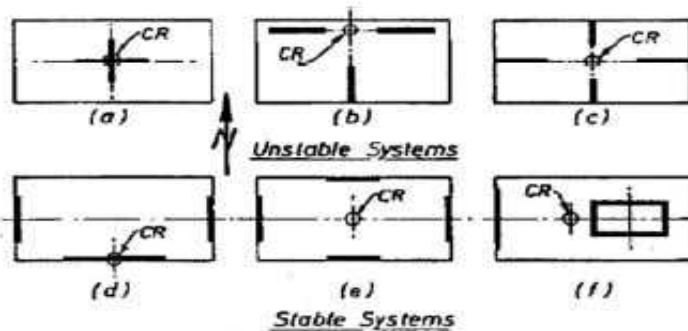
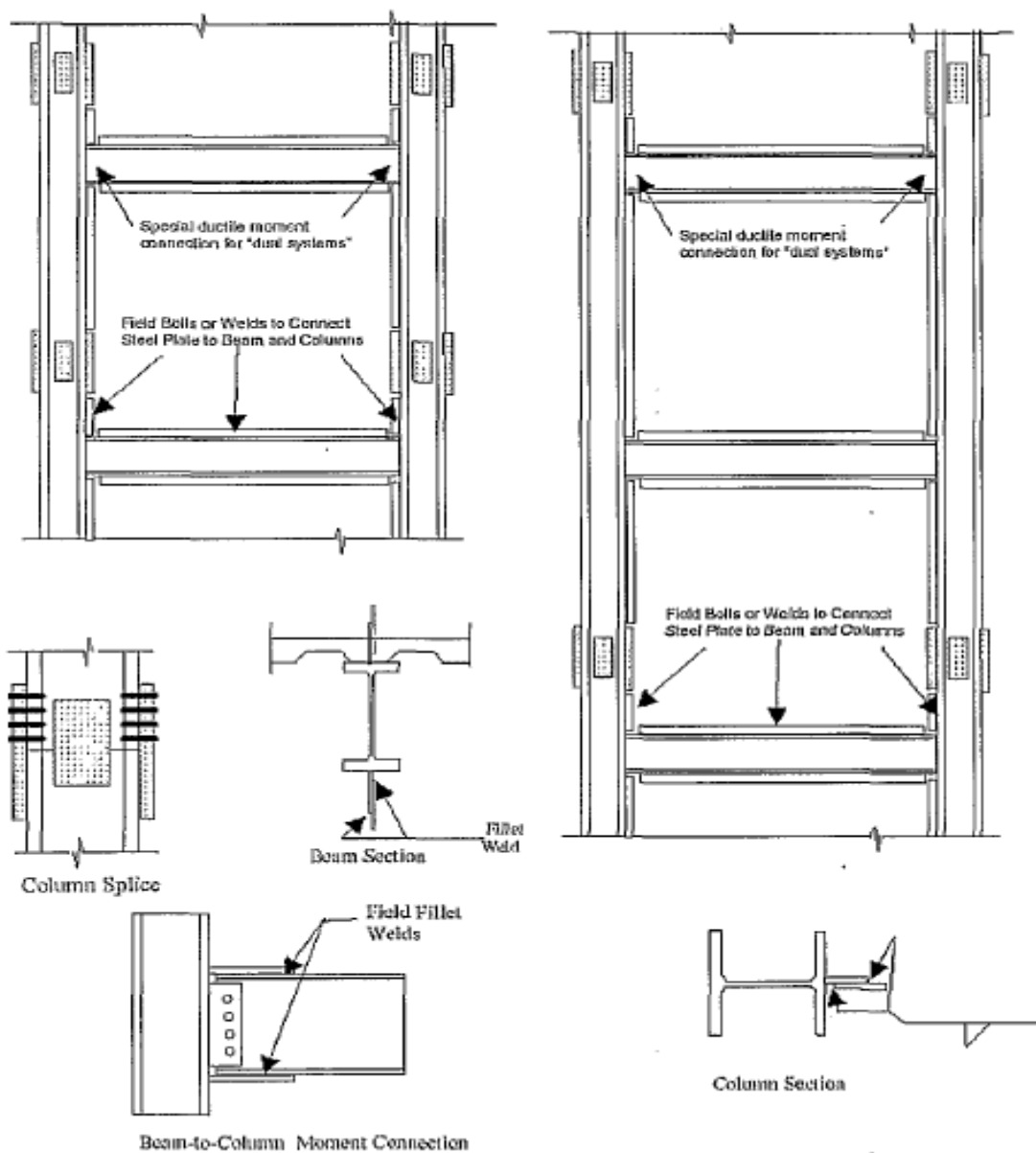


Coupled Shear Walls



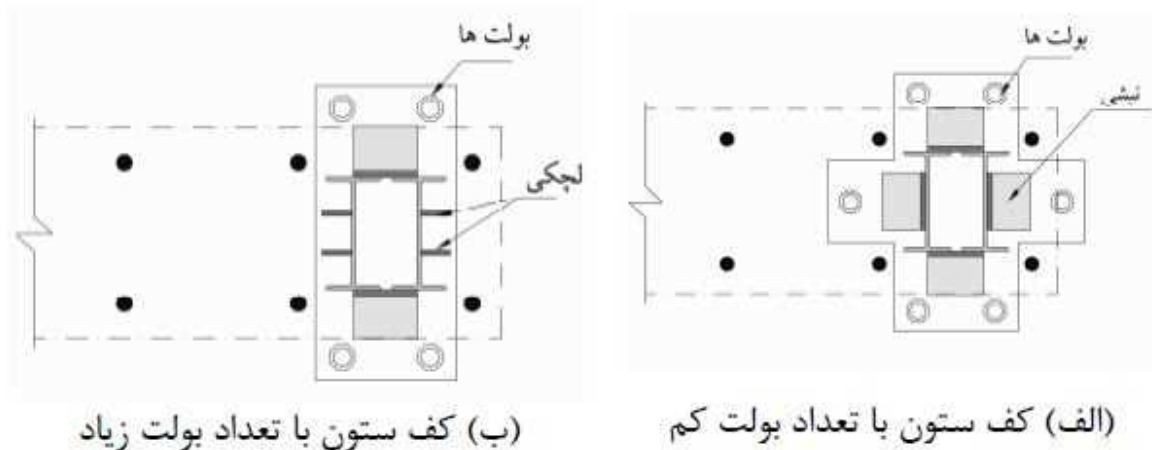
Out-ripper Shear





شکل‌های مرسوم برای دیوار برشی

حالت‌های ناپایدار و پایدار پیش‌بینی برای دیوار برشی



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